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The Solar and Geomagnetic Storms in 2024 May: A Flash Data Report

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Abstract

In 2024 May, the scientific community observed intense solar eruptions that resulted in a great geomagnetic storm and auroral extensions, highlighting the need to document and quantify these events. This study mainly focuses on their quantification. The source active region (AR; NOAA Active Region 13664) evolved from 113 to 2761 millionths of the solar hemisphere between May 4 and 14. NOAA AR 13664's magnetic free energy surpassed 10^{33} erg on May 7, triggering 12 X-class flares on May 8–15. Multiple interplanetary coronal mass ejections (ICMEs) were produced from this AR, accelerating solar energetic particles toward Earth. According to satellite and interplanetary scintillation data, at least four ICMEs erupted from AR 13664, eventually overcoming and combining each other. The shock arrival at 17:05 UT on May 10 significantly compressed the magnetosphere down to ≈5.04 R_E and triggered a deep Forbush Decrease. GOES satellite data and ground-based neutron monitors confirmed a ground-level enhancement from 2 UT to 10 UT on 2024 May 11. The ICMEs induced exceptional geomagnetic storms, peaking at a provisional Dst index of −412 nT at 2 UT on May 11, marking the sixth-largest storm since 1957. The AE and AL indices showed great auroral extensions that located the AE/AL stations into the polar cap. We gathered auroral records at that time and reconstructed the equatorward boundary of the visual auroral oval to 29°.8 invariant latitude. We compared naked-eye and camera auroral visibility, providing critical caveats on their difference. We also confirmed global disturbances of the storm-enhanced density of the ionosphere.

Unified Astronomy Thesaurus concepts: Solar storm (1526); Solar flares (1496); Solar coronal mass ejections (310); Solar active regions (1974); Sunspot groups (1651); Cosmic rays (329); Solar particle emission (1517); Geomagnetic fields (646); Aurorae (2192); Earth ionosphere (860); Planetary magnetospheres (997)

1. Introduction

Sunspot active regions occasionally direct interplanetary coronal mass ejections (ICMEs) to the Earth, disrupting near-Earth space environments, altering cosmic ray radiation levels, perturbing terrestrial magnetic fields, and extending the auroral oval equatorward (T. Pulkkinen 2007; M. Temmer 2022; E. W. Cliver et al. 2022; K. Kusano 2023; I. G. Usoskin 2023). Empirical evidence suggests that such solar eruptions occur more often in larger solar cycles, especially in their maximum and declining phase, although the maximum magnitude of the solar eruption is not necessarily dependent on the amplitude of the host solar cycle (E. K. J. Kilpua et al. 2015; L. Lefèvre et al. 2016; S. C. Chapman et al. 2020; M. J. Owens et al. 2022; S. C. Chapman & T. Dudok de Wit 2024). Such solar eruptions occasionally cause significant terrestrial impacts, such as great geomagnetic storms and increases of charged particle flux throughout the atmospheric profile, leading to an increased count rate of ground-based particle radiation detectors. These increases are known as ground-level enhancements (GLEs; E. W. Cliver et al. 2022; I. Usoskin et al. 2023). Extending analyses to the extremity of solar eruptions and their terrestrial consequences is essential to bridge modern scientific data with past supersized events, develop statistical analyses, and understand impacts on terrestrial infrastructure (I. A. Daglis 2001; D. N. Baker et al. 2008;

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C. D. Beggan et al. 2013; P. Riley et al. 2018; F. Miyake et al. 2019; R. T. Desai et al. 2020; M. Hapgood et al. 2021; K. Kusano 2023; I. Usoskin et al. 2023).

In this context, the last solar cycle, Solar Cycle 24 (from 2008 December to 2019 December), was notably quiet in terms of disturbances in the solar–terrestrial environments (S. Watari 2017; V. Manu et al. 2022). This solar cycle marked the lowest amplitude in sunspot number in the the 20th to 21st centuries since Solar Cycle 14 in 1902 January–1913 July (F. Clette et al. 2023). Solar Cycle 24 witnessed up to X14.8 in 2017 September in the revised scale (Table 7 of H. Hudson et al. 2024), geomagnetic storms reached a minimum Dst = −234 nT in 2015 March (N. Gopalswamy et al. 2015, 2024; Y. D. Liu et al. 2015; World Data Centre for Geomagnetism at Kyoto et al. 2015a; M. Piersanti et al. 2017; D. Webb & N. Nitta 2017), and hosted only two GLEs in 2012 May and 2017 September (A. L. Mishev et al. 2014, 2018; I. Usoskin et al. 2020).

The current solar cycle, Solar Cycle 25, commenced in 2019 December. Predictions suggest that this cycle may be either as quiet as or slightly more active than Solar Cycle 24 (P. Bhowmik & D. Nandy 2018; W. D. Pesnell & K. H. Schatten 2018; S. W. McIntosh et al. 2020; K. Mursula 2023; L. A. Upton & D. H. Hathaway 2023; F. Clette et al. 2024), whereas lessons from the previous cycle caution us to closely monitor how these forecasts align with the actual temporal evolution of the sunspot number (W. D. Pesnell 2016, 2020). Despite these predictions, Solar Cycle 25 has shown increased activity compared to Solar Cycle 24 in terms of disturbances in terrestrial environments (D. Oliveira et al. 2024). Even prior to reaching its maximum, Solar Cycle 25 has already witnessed one severe geomagnetic storm (with a minimum Dst below −200 nT according to the classic classification of C. A. Loewe & G. W. Prölss 1997) in 2023 April, reaching a minimum Dst of −213 nT (L. Li & S. Jin 2024), and one GLE in 2021 October (A. L. Mishev et al. 2022; A. Papaioannou et al. 2022; K.-L. Klein et al. 2022).

In 2024 May, Solar Cycle 25 witnessed the emergence of a significant sunspot active region, triggering multiple solar eruptions and leading to a GLE and a substantial geomagnetic storm. Notably, this geomagnetic disturbance reached −412 nT in the provisional Dst index, which is classified as a "great" storm (min Dst < −350 nT) according to C. A. Loewe & G. W. Prölss (1997) and ranked as the sixth greatest in the Dst index starting from 1957 and the ninth greatest in the Dxt index starting from 1932 (A. Karinen & K. Mursula 2005; WDC for Geomagnetism at Kyoto et al. 2015a; K. Mursula et al. 2022). This geomagnetic storm extended auroral displays to lower magnetic latitudes (e.g., S. Ghosh 2024; M. Greshko 2024; D. K. Karan et al. 2024). This solar and geomagnetic storm has attracted the scientific community to start analyses on several aspects such as their solar radio bursts (O. Kruparova et al. 2024), geomagnetic impacts in the Kp index and HPO index (Y. Yamazaki et al. 2024), local geomagnetic and ionospheric impacts around Italy (L. Spogli et al. 2024), auroral imageries, and resultant satellite drags (W. E. Parker & R. Linares 2024). Obviously there are more to come.

This solar–terrestrial storm serves as a valuable reference point for connecting modern scientific data with past great storms, particularly regarding geomagnetic impacts. Therefore, it is valuable to have a flash global overview of this solar and geomagnetic storm on the basis of what is available in the immediate aftermath. Here, we quantify observational data sets from this solar–terrestrial storm to facilitate further analysis and discussions regarding this exceptional event and to contextualize it with historical great storms.

2. The Source Solar Eruptions

This event occurred in an ascending phase—or near the maximum—of SC25. The daily sunspot number (F. Clette et al. 2023) significantly increased on May 2, reaching 187 on May 6 and 186 on May 12. At the time of writing, some predictions place the expected maximum from late 2024 to early 2025 (S. W. McIntosh et al. 2020; L. A. Upton & D. H. Hathaway 2023; F. Clette et al. 2024).

A new active region (AR), known as NOAA Active Region 13664 according to the NOAA/SWPC region number, came into the southeastern limb of the visible side of the solar disk on May 1. Its temporal and spatial evolutions were documented in sunspot drawings and images in several observatories, such as the Royal Observatory of Belgium using the Uccle Solar Equatorial table (USET) telescopes (S. Bechet & F. Clette 2022). This sunspot group started as a bipolar group, with two penumbrae and some intermediate sunspots in between. This active region developed rapidly from May 3–7 and crossed the central meridian on May 7. The area measured from the USET sunspot drawing with the Digi Sun software (e.g., F. Clette 2011; H. Hayakawa et al. 2023a) was 1312 millionths of the visible solar hemisphere (hereafter MSH). This group was still growing and developed into a Fkc McIntosh type, consisting of several moderately sized penumbral sunspots. This group continued to grow until May 9 up to an area of 2761 MSH, consisting of a compact penumbra containing most of the spots, as shown in the sunspot drawing and white-light image (Figure 1). The area remained relatively stable for a period before rotating over the limb on May 14. The evolution of the group area, as measured from sunspot drawings, is illustrated in Figure 2, showing a rapid increase until May 10.

The unsigned magnetic flux constantly increased from May 7 to 1.9×10^{23} Mx before it reached the west limb. We extrapolated the nonlinear force-free field (NLFFF; S. Inoue et al. 2014) of this AR using near-real-time data of the photospheric vector magnetic field from the Spaceweather HMI Active Region Patch (SHARP) near-real-time (NRT) data set (M. G. Bobra et al. 2014). Figure 2 shows the magnetic free energy, which is defined as the difference between the volume-integrated magnetic energy of the NLFFF and the potential magnetic field. We can see that the magnetic free energy increased before the M-class flares on May 2 and quickly grew on May 7 to $\approx 10^{33}$ erg before the consecutive X-class flares (Figure 2 and Table 1). The maximum values of the unsigned magnetic flux and magnetic free energy are as large as those of AR NOAA 12192, which provides the largest sunspot area, unsigned flux, and magnetic free energy in SC 24 (K. Karimov et al. 2024).

Figure 3 shows the distribution of the radial magnetic field and the contours of nonpotential magnetic field intensity $|\boldsymbol{B}\text{-}\boldsymbol{B}_{\rm p}| = 1000$ G (green surfaces), where $\boldsymbol{B}$ and $\boldsymbol{B}_{\rm p}$ are the magnetic fields from the SHARP-NRT data and the potential magnetic field. Red lines are the magnetic field lines of force for the nonlinear force-free field extrapolated by the magneto-hydrodynamic (MHD) relaxation method (S. Inoue et al. 2014). The notable feature of this region is the presence of several high free energy regions (HiFERs) where the nonpotential field intensity exceeds 1000 G. Likely, the large free energy and the

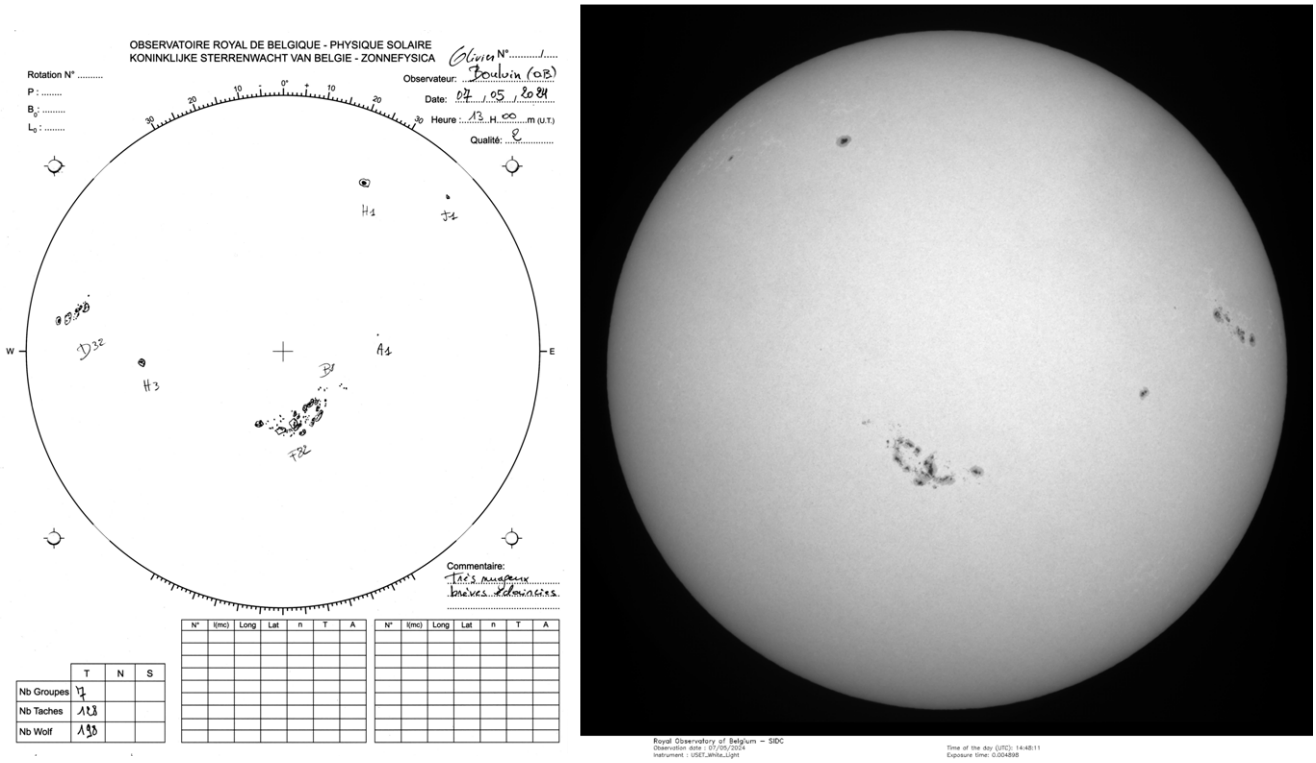


Figure 1. Sunspot drawing in a projected image and white-light image on 2024 May 7, as recorded with Uccle Solar Equatorial table telescopes in the Royal Observatory of Belgium (in a projected image). NOAA 13664 is located close to the central meridian.

complex structure of the HiFERs distribution may cause consecutive large flares (K. Kusano et al. 2020).

Eventually, this active region hosted 12 X-class flares from May 1 to 15 (Table 1 and Figure 2). Most of these were produced during transit over the visible solar hemisphere, while four were produced after the AR went over the western limb onto the far side. This AR produced a flare as great as X8.7. Even though the flare partially occurred on the western limb, this was the largest solar flare in the current solar cycle (SC25) at the time of writing at that time. Obviously there is more to come in this solar cycle, as exemplified by the X9.0 flare from NOAA AR 13842 on October 3. Figure 2 shows the soft X-ray flux continuously above the M1 level from May 8 to 12.

3. Coronal Mass Ejections

These solar eruptions of 2024 May released a considerable number of CMEs. From 2024 May 5 to 15, the Large Angle Spectroscopic Coronagraph (LASCO) observed at least 19 major CMEs (width $\geqslant 60°$), including 10 halo CMEs, which were produced by NOAA AR 13664 according to the data set being prepared by Gopalswamy's team. The calibrated LASCO data became available in 2024 August in the Coordinated Data Analysis Workshop (CDAW) for creating the CME Catalog (S. Yashiro et al. 2004, N. Gopalswamy 2009).

We have identified the CME that was most likely responsible for inducing the space environment disturbances observed near the Earth on May 11 with one (X1.0 at 22:36 UT) on May 8 that occurred in quick succession. The Solar Dynamics Observatory/Atmospheric Imaging Assembly (SDO/AIA; J. R. Lemen et al. 2012) observed the eruptive features in both flares. However, only a single halo CME was observed at 22:36 UT (#2 in Table 1, hereafter referred to as CME2), as shown in Figure 4. The two eruptions observed in the AIA likely merged before reaching the field of view of LASCO C2. The projected speed of CME2 was 952 km s^{-1} and if it traveled at this speed without any deceleration/acceleration, it would have reached Earth at ≈17:33 UT on May 10 (Table 1). While this is slightly later than the actual CME arrival of 17:05 UT on May 10, this offset shows a considerably good agreement in contrast with the usual error margins (up to ±8 to ±12 hr) of the CME arrival time predictions (N. Gopalswamy et al. 2001, 2005, Y. Zhong et al. 2024). In several cases, CMEs propagate earthward faster than expected from their apparent propagation speed measured in coronagraphs. In such cases, the actual arrival time comes earlier than the estimated arrival time. Based on these findings, we believe that CME2 was associated with the sudden storm commencement (SSC) at 17:05 UT on May 10. This estimated arrival time indicated that CME2 likely stacked on CME1 (#1 in Table 1). Additionally, two more halo CMEs were observed, which were associated with M-class flares (Table 2) produced by NOAA AR 13664. CME2 likely collided and stacked upon the three preceding CMEs originating from the same AR, resulting in a complex CME structure, as discussed later. Therefore, the great geomagnetic storm of 2024 May 10–11 is not associated with a single fast CME but with a complex accumulation of multiple CMEs and southward interplanetary magnetic fields (IMFs).

Subsequently, the AR produced more halo CMEs upon those, including two major flares (X1.1 at 09:24 on May 9 and X3.8 at 07:12 on May 10) launched CMEs at similar speeds (1226 and 1006 km s^{-1}) and another (X5.8 at 01:36 on May 11) launched a faster CME (1512 km s^{-1}), as shown in Figure 4. These CMEs probably arrived on Earth at ≈9:00 UT on May 12. This small difference between the observed and estimated arrival times, as indicated in Table 1, may imply that the preceding CME sequence cleared the interplanetary space,

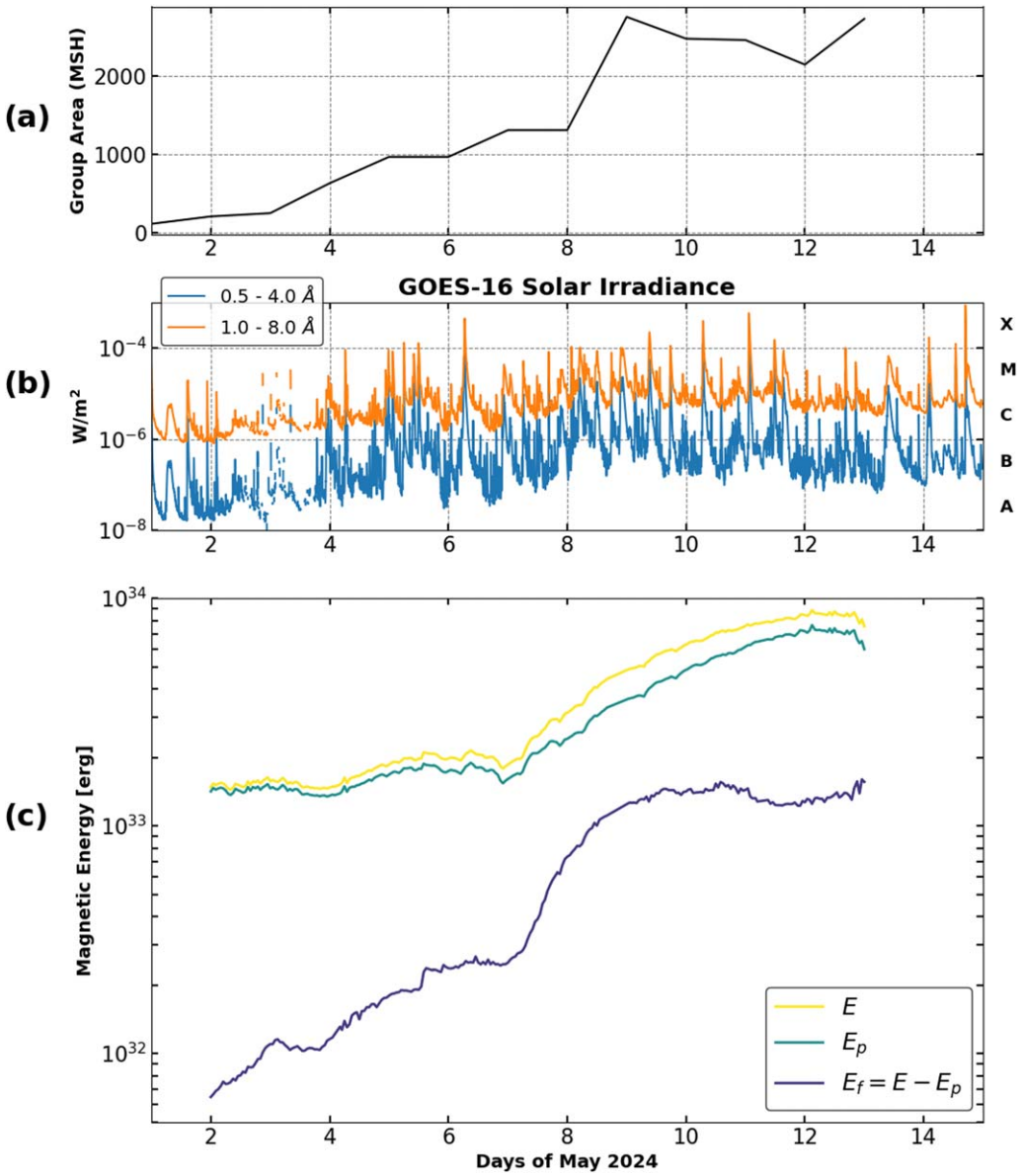


Figure 2. Temporal evolutions of the NOAA AR 13664 in millionths of the visible solar hemisphere (MSH) based on the USET measurements, GOES-16 X-ray fluence in 0.5–4.0 Å and 1.0–8.0 Å, as derived from the NOAA web portal, and the magnetic energy of the active region NOAA 13664 as a function of the day of 2024 May. E, E_p, and E_{free} are the total energy of the nonlinear force-free field, the potential magnetic field, and the free magnetic energy (E–E_p).

thereby allowing subsequent CMEs to propagate without significant deceleration (Y. D. Liu et al. 2014, 2019; D. Shiota & R. Kataoka 2016; M. Temmer et al. 2017; R. T. Desai et al. 2020). Additionally, the solar wind remained compressed owing to these CMES, resulting in a prolonged geomagnetic storm.

CMEs are typically faster or slower than their measured projected velocity, owing to the propagation process in the interplanetary space and their actual 3D structure. This is because the measured projected velocity, which is based on transverse expansion speed, is not necessarily equal to the propagation speed in the Sun–Earth direction. The difference in the speeds depends on the shape of the CME. Therefore, the actual arrival time mostly varies from the estimated arrival time, as indicated in Table 1. Furthermore, NOAA AR 13664 produced multiple CMEs at significantly different speeds over several days. Therefore, the relationship between the CMEs observed through coronagraphs and the CME structures observed in situ can be considerably more complex. To interpret this relationship, magnetohydrodynamic (MHD) simulations of multiple CME propagations should be further analyzed. Our results are considerably different from Spogli et al. (2024), who used the CME speeds automatically obtained from CACTus (E. Robbrecht & D. Berghmans 2004; E. Robbrecht et al. 2009) and associated the solar eruptions on May 9–11 with this great geomagnetic storm. The CACTus median speeds are generally lower than manual measurements, and their maximum speeds are occasionally extremely high (e.g., Figure 14 of E. Robbrecht and D. Berghmans 2004). Owing to their complexity, in-depth analyses of the interplay between CMEs are currently underway by Daikou Shiota's team.

Radio emissions from sources from the outside of the solar system can be scattered when CMEs cross the line of sight during observation and are detected as interplanetary scintillations (IPS). IPS observations have been used to study CMEs in the interplanetary space between the Solar and Heliospheric

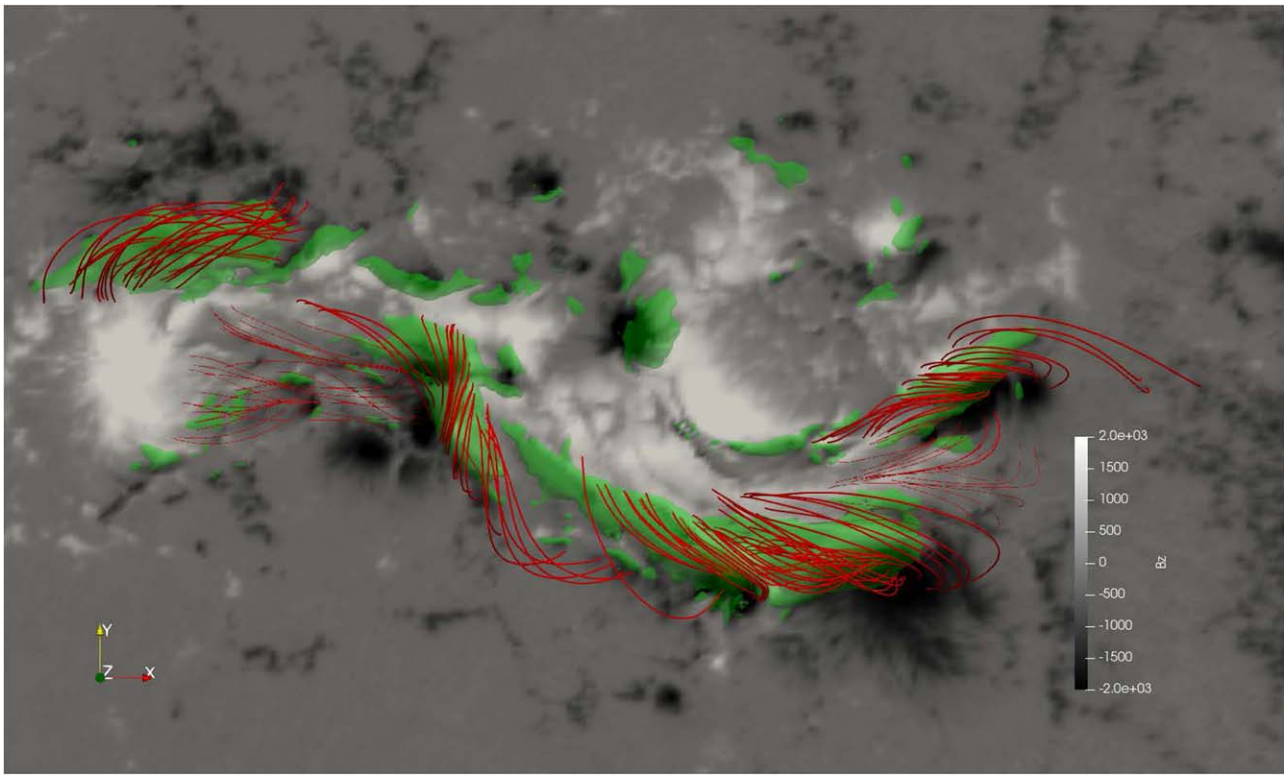


Figure 3. The distribution of the magnetic field of NOAA AR 13664 at 06:00 UT on 2024 May 10, just before the X3.9 flare. The gray-scale background indicates the radial magnetic field (B_z [G]), and the green surface is the contour of 10^3 G for the nonpotential field intensity, corresponding to HiFERs. Red lines show selected magnetic field lines of the nonlinear force-free field.

Table 1
Summary of the CMEs from NOAA AR 13664 Based on the SOHO/LASCO Data Based on their Calibration in 2024 August

ID	GOES Peak Time	GOES Class	Location	CME Speed	Travel Time*	Arrival Time*
1	2024-05-08 05:09	X1.0	S22W11	530	78:49	05–11 11:58
2	2024-05-08 21:40	X1.0	S20W17	952	43:53	05–10 17:33
3	2024-05-09 09:13	X2.2	S20W24	1280	32:38	05–10 17:51
4	2024-05-09 17:44	X1.1	S17W28	1024	40:48	05–11 10:32
5	2024-05-10 06:54	X3.9	S17W34	953	41:31	05–12 00:25
6	2024-05-11 01:23	X5.8	S17W44	1614	25:53	05–12 03:16
7	2024-05-11 11:44	X1.5	S19W60	No CME	…	…
8	2024-05-12 16:26	X1.0	S20W75	No CME	…	…
9	2024-05-14 02:09	X1.7	S17 WL	805	…	…
10	2024-05-14 12:55	X1.2	S17 WL	749	…	…
11	2024-05-14 16:51	X8.7	S18 WL	2010	…	…
12	2024-05-15 08:37	X3.4	S18 WL	1648	…	…

Note. The flare magnitudes and source locations are derived from the Event Archives of Solar Software of the Lockheed Martin Solar & Astrophysics Laboratory, with a minor modification for the location of the limb eruptions. We have annotated the longitude of the limb flares on the western limb as WL, in contrast with the Event Archives of Solar Software. The CME (coronal mass ejection) speeds in km s^{-1} are derived from the Solar and Heliospheric Observatory/Large Angle Spectroscopic Coronagraph (SOHO/LASCO) data set that is in preparation under Nat Gopalswamy's team. Their travel time (in hh:mm) and arrival time are estimated under the assumption of no deceleration of the CME speed in the fifth column as measured in the SOHO/LASCO data. This is an unrealistic assumption with a large error margin, as CMEs do not slow down much, only in extreme preconditioning cases. We estimated travel time and arrival time for CMEs originating from the central meridian (<|45|°from the central meridian).

Observatory (SOHO) coverage and the geospace space satellites located around L1. In this study, we analyzed the IPS observational data from early May measured by the ISEE, Nagoya University, at 327 MHz (M. Tokumaru et al. 2011). Immediately before the shock reached Earth, an increasing number of large-amplitude IPS responses were observed from May 8 to 10, as shown in Figure 5. It is evident that large-amplitude IPS responses were detected across numerous radio sources, particularly in the line of sight in the direction of large solar separation angles, where these CMEs were expected to be situated. IPS response enhancements can be associated with high-density regions where fast-propagating CMEs accumulate background solar winds (K. Iwai et al. 2019, 2021). Multiple CME mergers can enhance the compression of the solar wind plasma (C. Scolini et al. 2020). Notably, G. J. Koehn et al. (2022) tried a parametric study on this phenomenon and

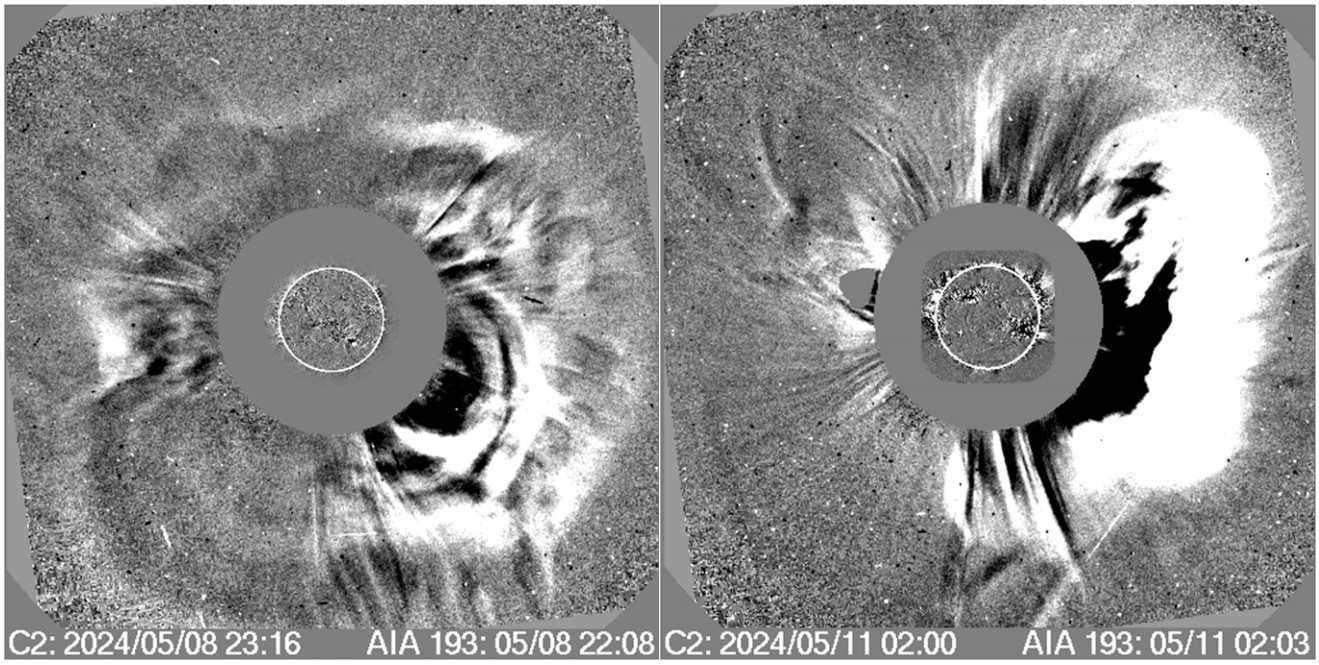


Figure 4. Examples of halo CMEs produced by NOAA AR 13664, as observed by LASCO C2 (G. E. Brueckner et al. 1995) from 22:08 to 23:16 UT on 2024 May 8 (left) and 02:00–02:03 UT on 2024 May 11 (right). Image courtesy of the CDAW of NASA.

Table 2
Major CMEs Produced by NOAA AR 13664, as Observed by the Large Angle Spectroscopic Coronagraph (LASCO)

Date	CME Start Time (UT)	CME Speed (km s^{-1})	CME WD (°)	Flare Start Time (UT)	Flare Class (GOES Class)	Location	NOAA
2024-05-05	15:36	353	157	14:12	C7.5	S20E19	13664
2024-05-07	04:48	311	56	02:59	C9.5	S21E05	13664
2024-05-08	02:24	358	169	02:16	M3.4	S17W04	13664
2024-05-08	05:36	530	360	04:37	X1.0	S22W11	13664
2024-05-08	12:24	677	360	11:26	M8.7	S20W17	13664
2024-05-08	22:36	952	360	21:08	X1.0	S20W17	13664
2024-05-09	09:24	1280	360	08:45	X2.2	S20W26	13664
2024-05-09	12:24	563	53	11:52	M3.1	S16W40	13664
2024-05-09	19:18	1024	360	17:23	X1.1	S14W28	13664
2024-05-10	07:12	953	360	06:27	X3.9	S17W34	13664
2024-05-11	01:36	1614	360	01:10	X5.8	S15W45	13664
2024-05-11	16:12	846	93	14:46	M8.8	S15W49	13664
2024-05-13	09:12	1690	360	08:48	M6.6	S20W81	13664
2024-05-14	01:48	331	63	01:23	M2.6	S17W88	13664
2024-05-14	02:24	805	172	02:03	X1.7	S19W88	13664
2024-05-14	13:00	749	78	12:40	X1.2	S17W92	13664
2024-05-14	17:00	2010	360	16:46	X8.7	S18W96	13664
2024-05-15	08:36	1648	360	08:13	X3.5	S18W98	13664
2024-05-15	10:48	1024	194	09:47	M3.6	WL	13664
2024-05-15	21:18	1355	360	20:30	C5.2	WL	13664

Note. These data were obtained from the data set being prepared by Nat Gopalswamy's team. The table presents the dates, start times in Universal Time (UT), speed, and angular width (WD) of the CMEs, as well as the flare start times, GOES X-ray classes, location, and NOAA AR numbers for the solar sources. The values were updated after a calibration conducted in 2024 August. We have annotated the limb eruption in the western limb as WL (western limb) to annotate their source location.

showed how two CMEs with velocities of 500 and 1500 km s^{-1} could combine to produce a great geomagnetic storm of minimum Dst of ≈ -400 nT (Figure 10 of G. J. Koehn et al. 2022). This is similar to what happened in 2024 May. Furthermore, CME compression can generate intense radio scattering if a region of concentrated density disturbance is created (Equation 1 of K. Iwai et al. 2019; A. T. Young 1971). This may support our aforementioned discussion of the CME–CME interactions. Further in-depth analyses are in preparation by Kazumasa Iwai's team.

4. Cosmic Ray Variations in Near-Earth Space and on the Ground

To monitor the cosmic ray intensity around this solar–terrestrial storm, we gathered proton flux data from the eastern

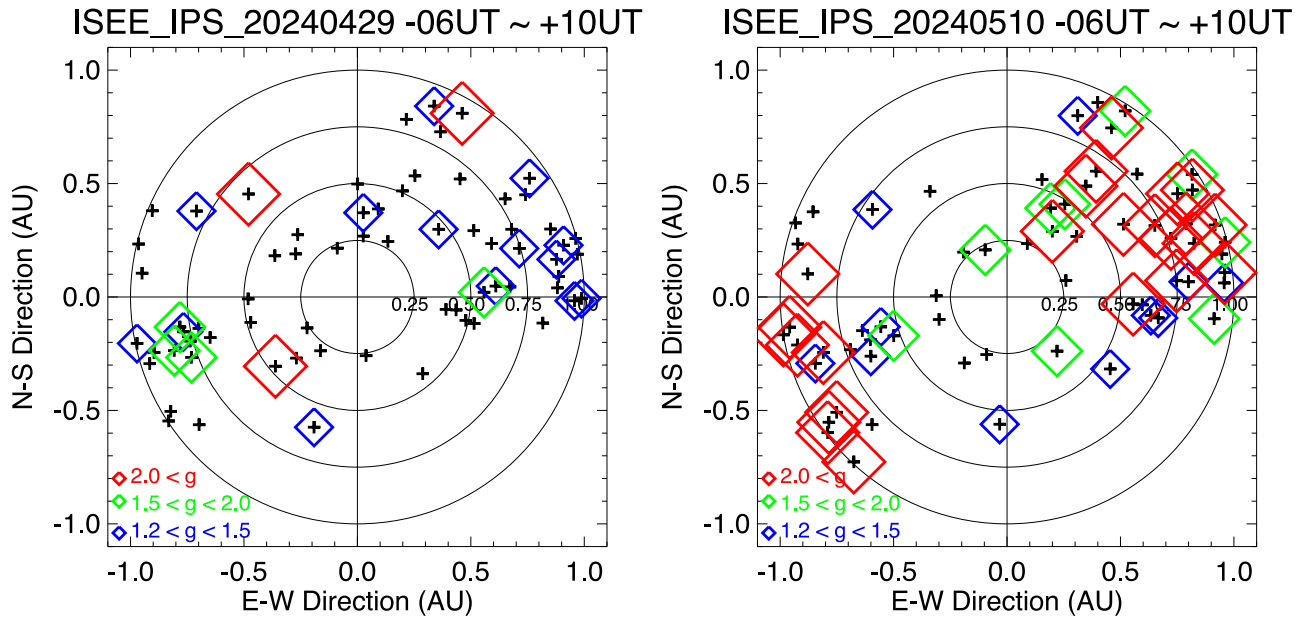


Figure 5. All-sky map of the IPS amplitude (g-value) observed by ISEE, Nagoya University on (a) 2024 April 29, when the NOAA AR 13663 was not on the disk, and (b) 2024 May 10. +: all observed radio sources, diamonds: radio sources with g-values as follows: $2.0 < g$ (red), $1.5 < g < 2.0$ (green), and $1.2 < g < 1.5$ (blue).

and western monitors of the Solar and Galactic Proton Sensor (SGPS) on board the GOES-16 satellite (B. T. Kress et al. 2021) downloaded from the NOAA data portal. We also acquired ground-based neutron monitor records from high-latitude regions, namely those of Oulu (Oulu in Finland), THUL (Thule in Greenland), SOPO (South Pole in Antarctica), and MWSN (Mawson in Antarctica), from the Oulu neutron monitor website and the Neutron Monitor Database, as summarized in the International GLE Database (I. Usoskin et al. 2020). These data are shown in Figure 6 with their source URLs.

During the period under investigation in this study, specifically on 2024 May 10–11, we observed a significant variation in cosmic ray activity. Corresponding to the arrival of CMEs at 17:05 UT, as shown by the SSC in the Kakioka Observatory (Kakioka Magnetic Observatory et al. 2013; see details in Section 6), a significant Forbush Decrease (S. E. Forbush 1937) in cosmic ray intensities was observed in all available data sets with different cutoff rigidities. The maximum decrease in the neutron monitor count rate of magnitude $\geqslant$15% or more was reached around midnight, after which the recovery phase began.

Around 2 UT on May 11, the flux of high-energy (>500 MeV) protons started to increase, as registered by the GOES-16 satellite (Figure 6). Notable variation of the solar proton flux triggered GLE alerts, e.g., an alert of the Bartol group, the Athens GLE alert ++ by the Athens group, and an alert issued at 01:55 UT by the Kazakhstan group (private communication with O. Kraykunova). However, the two latter alerts were produced using several midlatitude stations and are likely due to a reduction in the cutoff rigidity resulting from strong magnetospheric storms (K. Kudela et al. 2008). Detailed neutron monitor records are provided in the Neutron Monitor Database (H. Mavromichalaki et al. 2011) and the International GLE database (I. Usoskin et al. 2020). According to a careful inspection of the records, polar neutron monitors, such as the South Pole, Thule, Oulu, and Mawson (Figure 6), exhibit a small increase from $\approx$2 UT to 10 UT during the recovery of the deep Forbush Decrease, whose profile corresponds to the SEP profile registered by the SGPS on board the GOES-16 satellite. Since the cutoff rigidity of these neutron monitor stations is nearly zero, the magnetospheric effect, that is, the reduction of the cutoff rigidity, cannot be used to explain the count rate increase.

Therefore, according to the recently updated GLE definition,[18] the global neutron monitor network observed a GLE. Therefore, this event was confirmed as GLE #74, the second of the current Solar Cycle 25, after GLE #73 on 2021 October 28 (A. L. Mishev et al. 2022; A. Papaioannou et al. 2022). Accordingly, this GLE was included in the International GLE Database. As discussed above, the neutron monitor count rate increases during this event are a complicated interplay between the recovery of the Forbush Decrease, increased SEP flux, and geomagnetic storm impact; the former plays a role at polar stations and the latter at mid- and low-latitude stations. A detailed analysis of this complex event is desired and reserved for further dedicated studies.

5. Interplanetary Shock Arrivals and Magnetospheric Compressions

We used data sets of measurements conducted by the Time History of Events and Macroscale Interactions during Substorms (THEMIS) spacecraft (V. Angelopoulos 2008) to evaluate the magnetospheric compressions upon shock arrivals. We derived the magnetic field measurements from fluxgate magnetometers (U. Auster et al. 2008) and ion moments from an electrostatic analyzer (J. P. McFadden et al. 2008). When the interplanetary shock arrived on 2024 May 10, all three THEMIS satellites were aligned within the post-noon sector of the dayside magnetosphere during the inbound segment of their orbits.

Figure 7 shows the THEMIS-E measurements of the magnetic field and plasma from 17:04 to 17:07 UT on May 10. From top to bottom, it shows the magnetic field strength ($B_{\rm total}$), magnetic field components (B_x, B_y, B_z) in the Geocentric Solar Ecliptic (GSE)

[18] "A GLE event is registered when there are near-time coincident and statistically significant enhancements of the count rates of at least two differently located high-elevation neutron monitors and a corresponding enhancement in the proton flux measured by a space-borne instrument(s)" (for details see S. V. Poluianov et al. 2017, and the discussion therein).

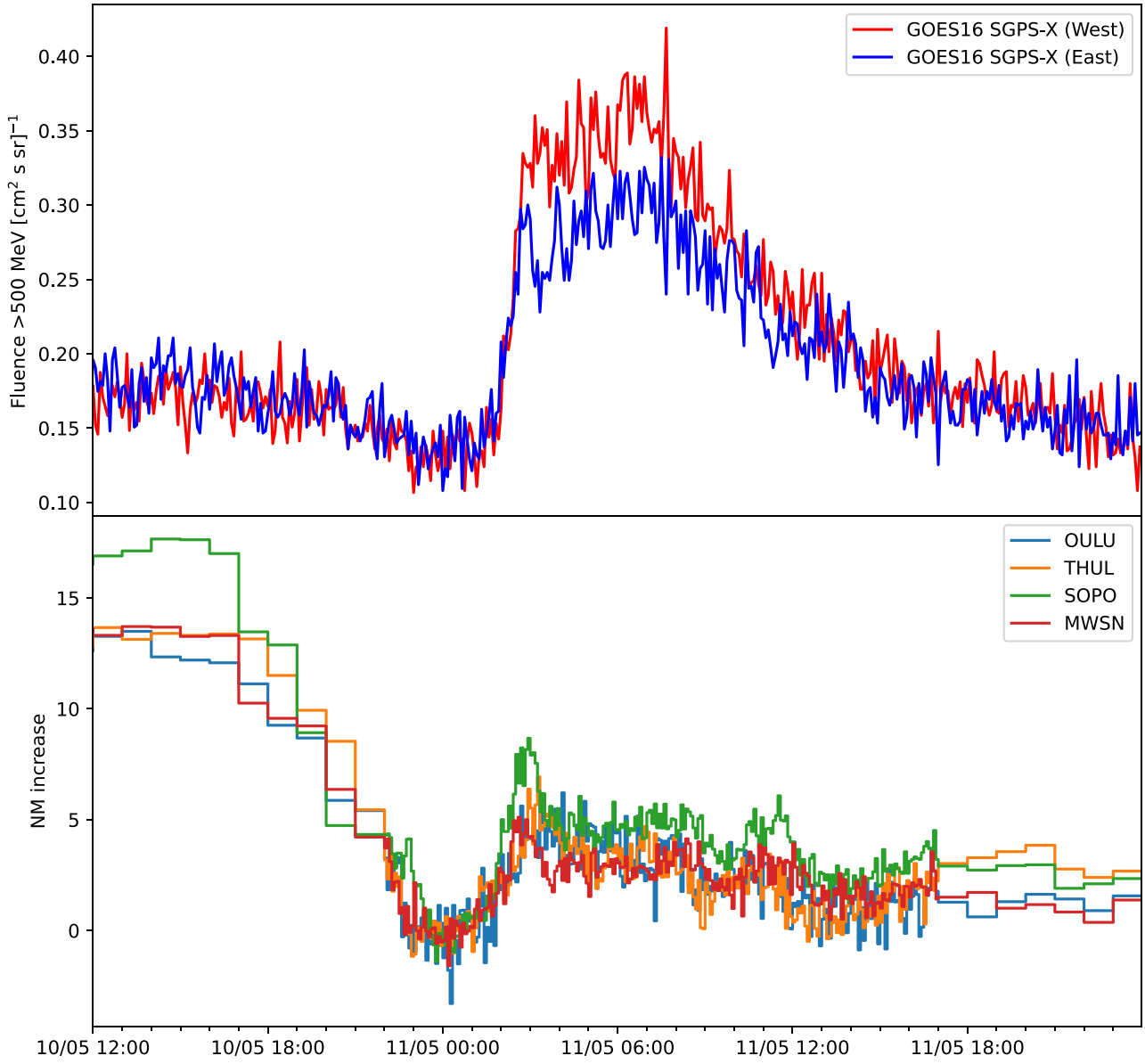


Figure 6. Cosmic ray flux >500 MeV observed by eastern- and western-facing monitors of the SGPS detector on board the GOES-16 satellite, based on the NOAA data portal, alongside the increase in ground-based neutron monitor count rates from high-latitude regions. The neutron monitor sites include Oulu (located in Finland at N65°03′, E025°28′, 15 m above sea level), THUL (located in Thule, Greenland, at N76°30′, W068°42′, 26 m above sea level), SOPO (located at the South Pole in Antarctica, at S90°00′, N/A, 2820 m above sea level), and MWSN (located at Mawson in Antarctica, at S67°36, E062°53′, 30 m above sea level), as sourced from the Oulu Neutron Monitor and Neutron Monitor Database, and summarized in the International GLE Database. All data are corrected for barometric pressures without a detrending procedure. For neutron monitor increase calculations, the baseline was selected from 23:00 on May 10 to 01:00 on May 11 for MWSN, SOPO, and THUL, and from 23:30 on May 10 to 01:30 on May 11 for Oulu.

coordinate system, ion energy flux spectrogram, bulk flow velocity (V_x, V_y, V_z) in the GSE coordinate system, and number density (N_i), respectively. The GSE spacecraft position is annotated in the lower part of the plot. Additionally, the vertical dashed line denotes the moment of the magnetopause crossing (17:05 UT) at $x = 7.1$ R_E, $y = 3.2$ R_E, and $z = -2.7$ R_E, indicating that THEMIS-E crossed over the magnetopause at ≈8.24 R_E. Figure 7 shows the sudden offsets of all magnetic field and plasma parameters, particularly the particle energy dispersion, at the magnetopause. Prior to the magnetopause, we can observe a period of a stable magnetic field with a positive B_z component accompanied by a low bulk flow velocity and high concentration of high-energy ions (>1 keV), which are typically observed in the Earth's magnetosphere. The region between the dashed and solid lines corresponds to the magnetosheath, which is characterized by a weaker magnetic field, broader ion energy spectrum, and high ion number density. The area after the solid line indicates the THEMIS-E transition into the solar wind, characterized by a high plasma flow velocity and narrow ion energy spectrum of approximately 1 keV.

Figure 8 shows the THEMIS-A measurements of the magnetic field and plasma from 17:04 to 17:07 UT on May 10. These data sets independently confirm the estimated values. The vertical dashed line denotes the moment of the magnetopause crossing (17:05 UT) at the [x, y, z] coordinates in the Earth's radius (R_E) of $x = 7.7$ R_E, $y = 2.4$ R_E, $z = -2.7$ R_E, indicating that THEMIS-A crosses over the magnetopause at ≈8.51 R_E, which is consistent with THEMIS-E's crossing at ≈8.24 R_E.

Figure 9 shows the THEMIS-E measurements of the magnetic field and plasma data in the same order as that in

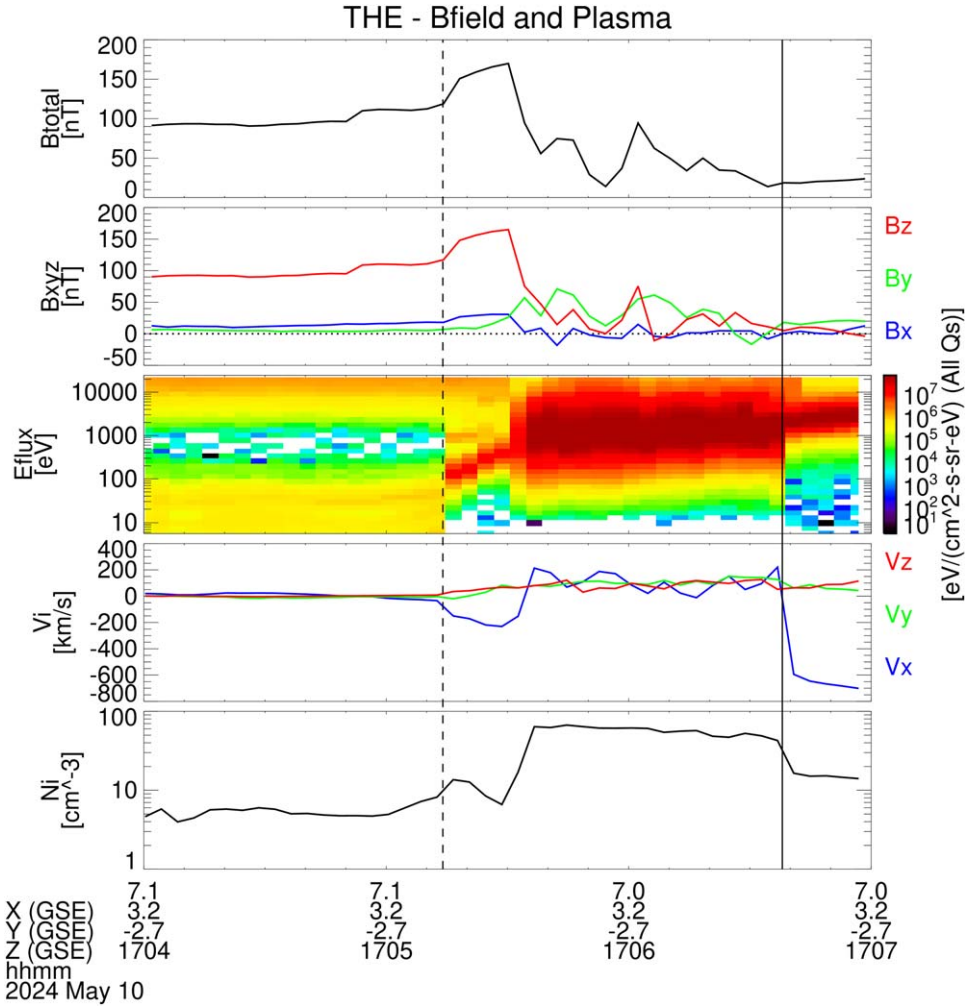


Figure 7. THEMIS-E's measurements of magnetic field strength ($B_{\rm total}$), magnetic field components (B_x, B_y, B_z) in the GSE coordinate system, ion energy flux spectrogram, bulk flow velocity (V_x, V_y, V_z) in the GSE, and number density (N_i) from 17:04 to 17:07 UT on May 10. The magnetopause crossing is indicated by a vertical dashed line.

Figure 7. In this case, the spacecraft crosses from the magnetosheath to the magnetosphere, encountering the magnetopause at $x = 3.4$ $R_{\rm E}$, $y = 3.2$ $R_{\rm E}$, and $z = -1.9$ $R_{\rm E}$ at approximately 19:12 UT, as denoted by the dashed line. This indicates that THEMIS-E traverses the magnetopause at ≈5.04 $R_{\rm E}$. Initially, the magnetic field exhibits an intense negative B_z component, which subsequently undergoes significant rotation within the magnetosheath region. The ion bulk flow velocity is also high and fluctuates substantially in this region. A wider ion energy spectrum is discernible here. Upon entering the magnetospheric region, the magnetic field was still strong but devoid of high-frequency disturbances. Additionally, as expected, the ion bulk flow velocity of magnetospheric plasma was considerably lower, while the number density was higher than normal.

This compressed magnetopause value (≈5.04 $R_{\rm E}$) is somewhat extreme and lower than that of the geosynchronous orbit (6.6 $R_{\rm E}$). Figure 10 shows the THEMIS-A measurements of the magnetic field and plasma from 18:50 to 19:30 UT on May 10. The vertical dashed line denotes the moment of magnetopause crossing (19:19 UT) at $x = 4.0$ $R_{\rm E}$, $y = 2.9$ $R_{\rm E}$, and $z = -2.1$ $R_{\rm E}$, indicating that THEMIS-A crossed over the magnetopause at ≈5.37 $R_{\rm E}$. This is broadly consistent with THEMIS-E's crossing at ≈5.04 $R_{\rm E}$. Here, we considered entry into the magnetosphere to occur when the energy of the main particle population exceeds 1 keV and particles from the magnetosheath are absent, which are typical characteristics of the magnetosphere. The period preceding the dashed lines represents particles with a wide range of energies, and indicates a mixture of the magnetosphere and magnetosheath plasma. This region can be identified as a boundary layer characterized by open magnetic field lines. Overall, these measurements confirmed that the magnetosphere compressed considerably immediately after the shock arrival at 17:05 UT. However, further assessments of this extreme magnetospheric compression are required.

6. Geomagnetic Disturbances

The geomagnetic measurements captured at least four SSCs in 2024 May. The Kakioka ΔH measurement recorded SSCs at ≈14:07 UT on May 2 (26 nT), 17:05 UT on May 10 (78 nT), 05:51 UT on May 16 (10 nT), and 13:26 UT on 17 (18 nT), respectively (Kakioka Magnetic Observatory et al. 2015). According to the real-time solar wind data acquired by ACE, a jump in solar wind speed and density arrived at the L1 Lagrange point at 16:36 UT on May 10. In the vicinity of the jump, the speed increased from approximately 460 to

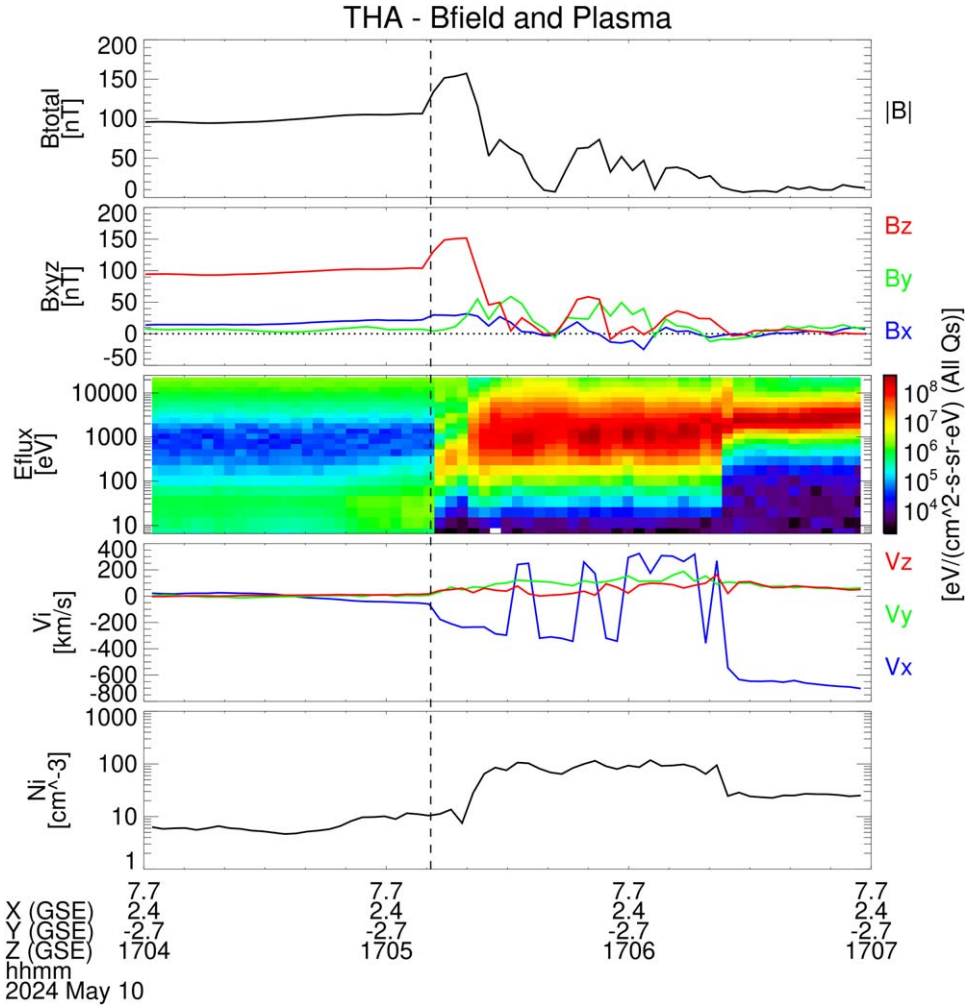


Figure 8. THEMIS-A's measurements of magnetic field strength ($B_{\rm total}$), magnetic field components (B_x, B_y, B_z) in the Geocentric Solar Ecliptic (GSE) coordinate system, ion energy flux spectrogram, bulk flow velocity (V_x, V_y, V_z) in GSE, and number density (N_i) from 17:04 to 17:07 UT on May 10. The magnetopause crossing is indicated by a vertical dashed line.

660 km s^{-1}, and the density increased from ≈10 to 27 cm^{-3}. This jump caused a substantial SSC from ≈17:05 UT on May 10 up to ≈78 nT in the Kakioka ΔH measurement (Kakioka Magnetic Observatory et al. 2015). This implies that the travel time from the L1 point to Earth is 29 min at the SSC moment. This estimate includes the transition time from the magnetopause to the ground.

Figure 11 summarizes ΔX observed at 20° < MLAT (magnetic latitude) < 30° around the SSC. Again, ΔX at 17:04 UT on 2024 May 10 (immediately before the SSC) is set to be zero as a baseline. An abrupt increase in ΔX is evident at 17:05–17:06 UT. The SSC amplitude of the abrupt increase in ΔX is larger on the duskside than on the dawnside. This is consistent with the previous studies and attributed to the contribution of the magnetospheric current and the ionospheric current as well (e.g., T. Araki et al. 2006). The European sector witnessed a relatively large SSC amplitude, as it was located in the duskside in the summer hemisphere upon the SSC. This favorable location probably enhanced ionospheric conductivity and the field-aligned current to increase the magnetic variations in the European sector (A. Shinbori et al. 2009, 2012). The amplitude is ≈130 nT, which is large but not unique in comparison with the historical records of SSC (T. Araki 2014; H. Hayakawa et al. 2022).

Figure 12 summarizes measurements of interplanetary magnetic field (IMF) and solar wind parameters, the SME indices, the AE and AL indices, the Dst and Dxt global indices, the Dxt local indices, and the absolute values of the magnetic latitudes at which the aurorae and stable red auroral (SAR) arcs were reported. The IMF and solar wind parameters are acquired from the 1 minute OMNI data (N. E. Papitashvili & J. H. King 2020). We note that the IMF and the solar wind parameters were observed by the ACE spacecraft at 245 $R_{\rm E}$ upstream of the solar wind, and the data were time shifted to the subsolar point of Earth's bow shock. Thus, the time variations are not always correct due to the ambiguity of the time-shifting algorithm, in particular during the highly disturbed conditions. Because of this reason, we intend to overview the IMF and solar wind parameters briefly, without going into detail, although the IMF data sets probably indicate at least one compressed flux rope inside the sheath of the subsequent CME(s) and the density profile indicates arrivals of multiple sheaths.

The SSC (Figure 11) was caused by rapid increases in the solar wind velocity from ≈450 km s^{-1} to >700 km s^{-1} and the solar wind density from ≈17 cm^{-3} to >30 cm^{-3}. At the same time, the magnitude of the auroral electrojet increased, as indicated by the AE, AL, SMU, and SML indices. The IMF B_z

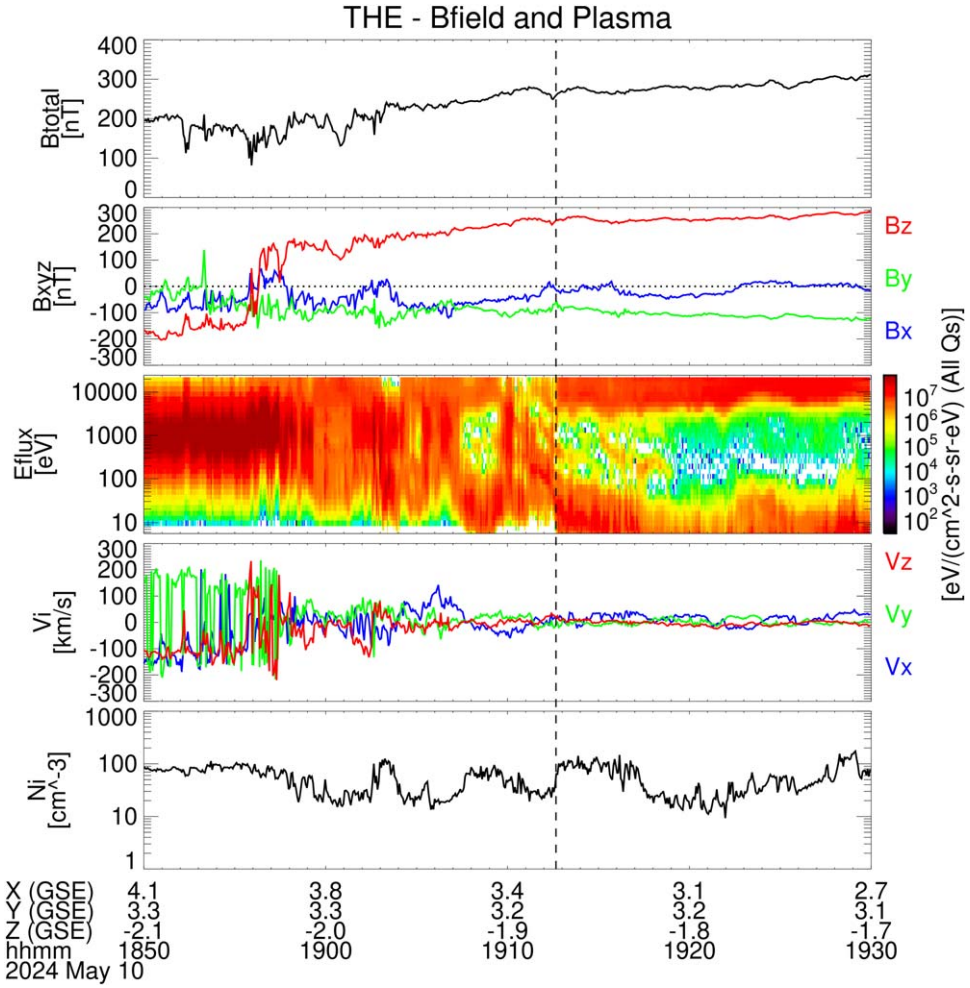


Figure 9. THEMIS-E's measurements of magnetic field strength ($B_{\rm total}$), magnetic field components (B_x, B_y, B_z) in the Geocentric Solar Ecliptic (GSE) coordinate system, ion energy flux spectrogram, bulk flow velocity (V_x, V_y, V_z) in GSE, and number density (N_i) from 18:50 to 19:30 UT on May 10. The magnetopause crossing is indicated by a vertical dashed line.

turned southward just after the SSC, reaching ≈ -40 nT around 18:08 UT. After the SSC, there are at least nine intervals in which IMF B_z remained negative for 30 minutes and more. They are summarized in Table 3. The solar wind speed gradually increased to 1026 km s^{-1} at ≈01:17 UT on May 12.

The Dst/Dxt indices rapidly decrease during the sustained southward IMF intervals of 1 and 2 listed in Table 3. They reached a minimum value of −412 nT/415.5 nT, respectively, during the sustained southward IMF interval 3. This marks the end of the storm main phase and the storm peak hour, and indicates a considerably fast decrease in Dst/Dxt with peak-to-peak amplitudes of 474 and 479.8 nT in 9 hr. According to K. Tsubouchi & Y. Omura (2007), it is possible to calculate a statistical return period of geomagnetic storms of this minimum value (min Dst = −412 nT) as 5.7 yr. The provisional Dst/Dxt values are subject to small changes until definite values are established (A. Karinen & K. Mursula 2005; WDC for Geomagnetism at Kyoto et al. 2015a; K. Mursula et al. 2023).

Figure 12 shows that the Dst and Dxt indices follow each other quite closely, with momentary differences varying between −9 nT and +6 nT. Over the depicted time interval, the Dxt index is only slightly (less than 1 nT) lower than the Dst index. These differences are due to the somewhat different normalizations of the two indices (K. Mursula et al. 2008). The multiple complicated structures of CMEs (Section 3) likely caused the two-step development of this storm. The intensification of the ring current was caused by the penetration of hot ions into the deep inner magnetosphere due to the dawn–dusk convection electric field intensified by the large-amplitude southward IMF and fast solar wind. During the main phase of the storm, the solar wind speed was as high as ≈700–800 km s^{-1}, and the southward component of the IMF was as high as ≈50 nT. The solar wind speed, maximum value of the southward component of the IMF, and minimum Dst index were similar to those of the large magnetic storm of 2003 November 20–21. The hot ions are speculated to penetrate as low as $L \approx 1.5$ to largely intensify the ring current (Y. Ebihara et al. 2005).

Figure 12 also shows the local Dxt indices at the four Dst/Dxt stations. Their average makes the Dxt index. The largest compression at 17 UT on May 10, close to the SSC occurrence, was observed at SJG (+93.9 nT), which was located at this time closest of all stations to noon, in the early afternoon of local time. The main phase development proceeded fairly similarly in all four stations (at different local times) until 22 UT on May 10, after which somewhat larger differences started appearing. Note that only one station, HER, reached its local Dxt minimum at 2 UT on May 11, at the time of the (global) Dst/Dxt minimum, while SJG had it 1 hr before, HON 2 hr

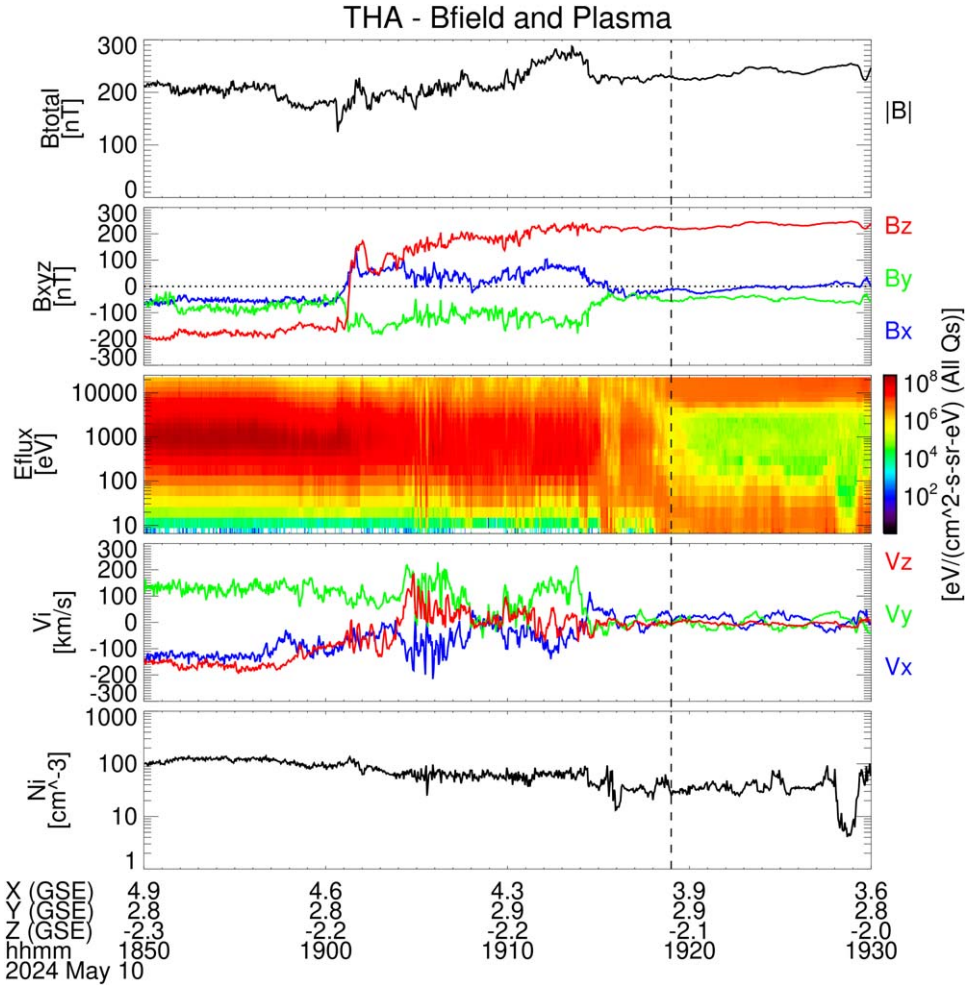


Figure 10. THEMIS-A's measurements of magnetic field strength (B_{total}), magnetic field components (B_x, B_y, B_z) in the Geocentric Solar Ecliptic (GSE) coordinate system, ion energy flux spectrogram, bulk flow velocity (V_x, V_y, V_z) in GSE, and number density (N_i) from 18:50 to 19:30 UT on May 10. The magnetopause crossing is indicated by a vertical dashed line.

afterward at 4 UT and KAK only at 8 UT, 6 hr after the Dxt minimum.

After 2 UT on May 11, the Dst/Dxt indices started increasing, and the storm recovery phase began. During the recovery, the Dst/Dxt indices showed some negative excursions corresponding to the southward turnings of the IMF B_z. Note that the local time of both KAK and HON at the time of their minimum observation was around 17 LT. For SJG it was 20 LT, and for HER it was 3 LT. This is in agreement with the fact that, statistically, at low latitudes, the largest disturbance is found around 18 LT (W. D. Cummings 1966; O. S. Yakovchouk et al. 2012). This local-time dependence of the storm-time geomagnetic disturbances is often thought to be the partial ring current but is probably not simple (N. Fukushima & Y. Kamide 1973). According to in situ observations, the peak of the ring current takes place in the dusk–midnight sector (N. Terada et al. 1998; Y. Ebihara et al. 2002; G. Le et al. 2004). This is different from the ground-based magnetic observations. This discrepancy is probably attributed to the contribution from the field-aligned current (S. Ohtani et al. 2010).

As shown in Figure 12, the auroral electrojet has also developed significantly. The provisional AE and AL indices (WDC for Geomagnetism at Kyoto et al. 2015b) initiated significant development at 17:06 UT on May 10 and increased to 2379 nT and −1830 nT, respectively, at 17:28 UT. These peak values were reached in 22 minutes. The abrupt jump in the provisional AE index was probably regarded as a shock-triggered substorm (S.-I. Akasofu & J. K. Chao 1980; B. T. Tsurutani & X.-Y. Zhou 2003). The ionospheric current associated with the main impulse of the SSC have also contributed to the abrupt jumps in these indices (T. Araki 1994; T. Zhang et al. 2023). During the storm's main phase, the provisional AE and AL indices reached peak values of 4098 and −3797 nT, respectively, at 19:48 UT on May 10. Subsequently, these indices remained within $<$ |2000| nT during the main storm phase. During the storm recovery phase, these indices reached peak values of 3982 and −3797 nT at 09:00 UT on May 11. According to a statistical study by M. Nakamura et al. (2015), up to 16 events occurred during which AL exceeded −3000 nT from 1996 to 2012, and the recurrence period for AL of −3797 nT was 1.49 yr. In this sense, the provisional AL index of −3797 nT was not that extreme. We have to note that the AE and AL indices are not necessarily proportional to the intensity of the auroral electrojet.

Based on the behaviors of the AE index and the AL index around the storm peak, the auroral oval is suspected to have extended significantly equatorward from the AE stations during this interval, resulting in the suppression or saturation of the AE and AL indices. The SME indices are used to overcome

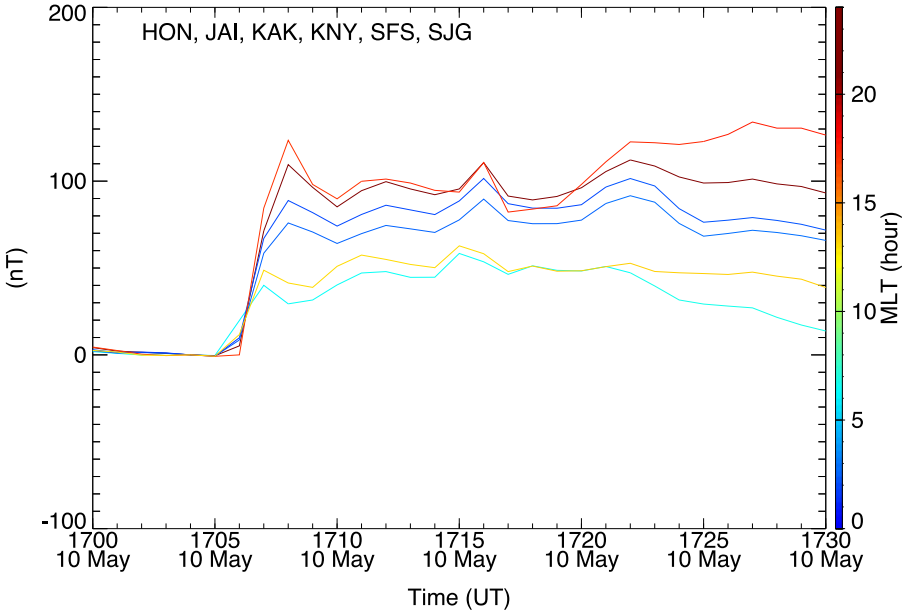


Figure 11. The north component (ΔX) of the geomagnetic disturbances at $20° < \mathrm{MLAT} < 30°$. The color codes indicate MLT (magnetic local time). The abbreviation codes indicate the sites used to plot.

this issue since it is calculated from many sites, usually exceeding 100 (P. T. Newell & J. W. Gjerloev 2011). In that sense, the SME indices are better. However, the AE index has been used to represent the magnitude of the auroral electrojet. We believe that AE is still useful to compare modern events with historical events and to drive empirical models depending on AE. In order to see this possibility, we have also checked the SME, including the SML index in Figure 12. The fourth panel of Figure 12 shows the SME indices. Interestingly, the peak amplitude of the SML index, which is an extension of the AL index, is not always larger than that of the AL index. These values are provisional, so a definitive conclusion regarding the auroral electrojet cannot be obtained.

Figure 13 summarizes the north component of the geomagnetic disturbances ΔX at different latitudinal zones. The data was provided by Intermagnet et al. (2021). The sites used to plot are summarized in Table 4. ΔX at 17:04 UT on 2024 May 10 is set to be zero. Diurnal variations, such as Sq, were not removed. Figure 13(a) shows ΔX at $\mathrm{MLAT} > 75°$. After the SSC, ΔX is negative on the duskside, whereas it is positive on the dawnside. The variations are reasonably explained by the twin vortices of the Hall current flowing in the ionosphere, known as the DP2 equivalent current (A. Nishida 1968). In the poleward part of the DP2 current, it tends to flow westwardly on the duskside and eastwardly on the dawnside, as schematically illustrated in the right-bottom corner of Figure 13. Figure 13(b) shows ΔX in the latitudinal zone, $60° < \mathrm{MLAT} < 70°$, which is usually referred to as the auroral zone. The lower envelope of ΔX in this panel is roughly equivalent to the AL index. The westward current is the most intense around midnight and in the postmidnight, which is consistent with the previous studies (e.g., Y. Kamide et al. 1998). In general, ΔX is positive on the duskside and negative on the dawnside. This is consistent with the expectation that ΔX is primarily, not all, caused by the equatorward part of the DP2 current. Figure 14(c) shows ΔX in the lower latitudinal zone, $20° < \mathrm{MLAT} < 30°$. The contribution from the ionospheric current is likely to be small. The contributions from the ring current, the tail current, the magnetopause current, and the field-aligned current are probably significant. The averaged ΔX should be close to the Dst index. During the main phase, the variations of ΔX show almost coherently with respect to MLT, as mentioned above, but it is clearly shown that ΔX tends to be smaller on the duskside than on the dawnside. After the storm peak, the MLT dependence became significant. There are some negative peaks at ≈02 UT, ≈08 UT, and ≈14 UT on May 11. These peaks took place in the premidnight sector, suggesting that they were primarily caused by the storm-time partial ring current. During the interval from ≈06 UT to ≈10 UT on May 11, ΔX shows a negative excursion on the premidnight sector, whereas it shows a positive excursion in the pre-noon sector. This can be reasonably explained using the development of the partial ring current in the premidnight sector due to the injection of fresh ions from the nightside plasma sheet and the reduction of the ring current in the pre-noon sector due to the escape of the preexisting ions in response to the rapid enhancement of the magnetospheric convection (P. C. Brandt et al. 2002; Y. Ebihara et al. 2002; K. K. Hashimoto et al. 2002).

7. Spatial Extents of the Auroral Visibility and the Auroral Oval: Naked Eye Observations

This storm significantly extended the auroral oval equatorward, as expected from the empirical correlation between the extension of the equatorward boundary of the auroral oval and the intensity of the associated geomagnetic storm (N. Yokoyama et al. 1998; S. P. Blake et al. 2021). As such, auroral visibility was reported worldwide. We followed a citizen science approach to gather auroral visibility reports. This is a valid approach for reconstructing the equatorward boundary of auroral visibility and auroral oval, as reported by N. A. Case et al. (2016). The authors prepared an online survey and called for auroral reports for the naked eye and instrumental visibility. This online inquiry allowed us to obtain 80 valid auroral reports, as exemplified in Figure 14. Figure 15 summarizes responses from this online survey. Our result is roughly consistent with independent survey

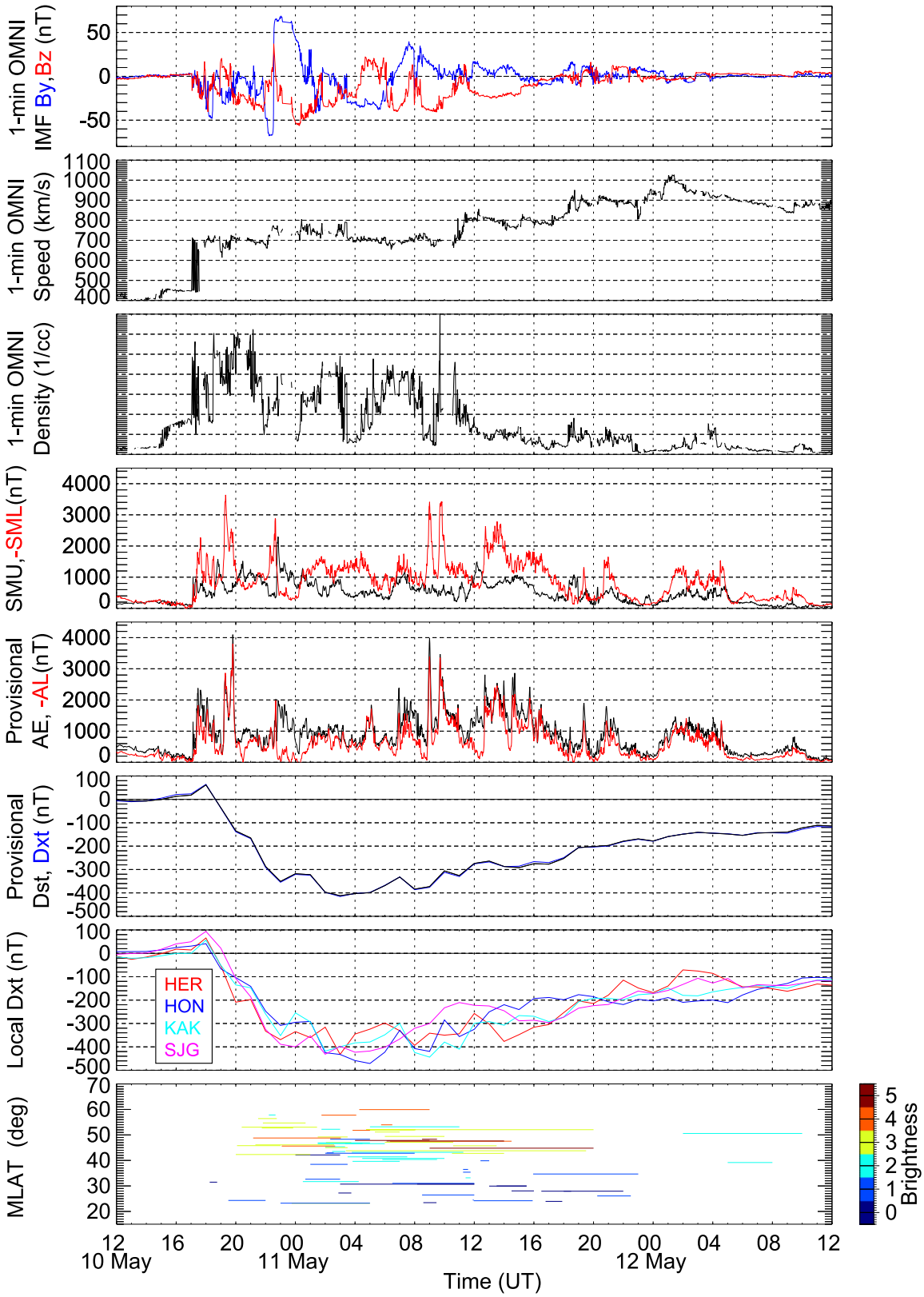


Figure 12. Temporal evolutions of interplanetary magnetic field (IMF), solar wind speed and density, geomagnetic disturbance, and auroral visibility. The first panel shows the interplanetary magnetic field of the solar wind in B_y (blue) and B_z (red) in the GSM coordinates. The second panel shows the solar wind velocity. The third panel shows the density of the solar wind. For these three panels, we have derived their source data from the 5 minute OMNI data. The IMF and the solar wind data are time shifted to the subsolar point of Earth's bow shock. Thus, the variations are subject to changes depending on the calculation algorithm. The fourth panel shows the SMU (black) and SML (red) indices. The fifth panel shows provisional AE (black) and AL (red) indices. The sixth panel indicates the (global) Dst and Dxt indices, and the seventh panel is the local Dxt indices at Hermanus (HER), Honolulu (HON), Kakioka (KAK), and San Juan (SJG). The last panel displays the absolute value of the magnetic latitude at which the aurorae and SAR arcs were reported both by naked eyes and instruments. The brightness is generally given with the International Brightness Coefficient (IBC classification), apart from 0 for brightness not enough for naked-eye visibility and 5 for brightness more than a full moon.

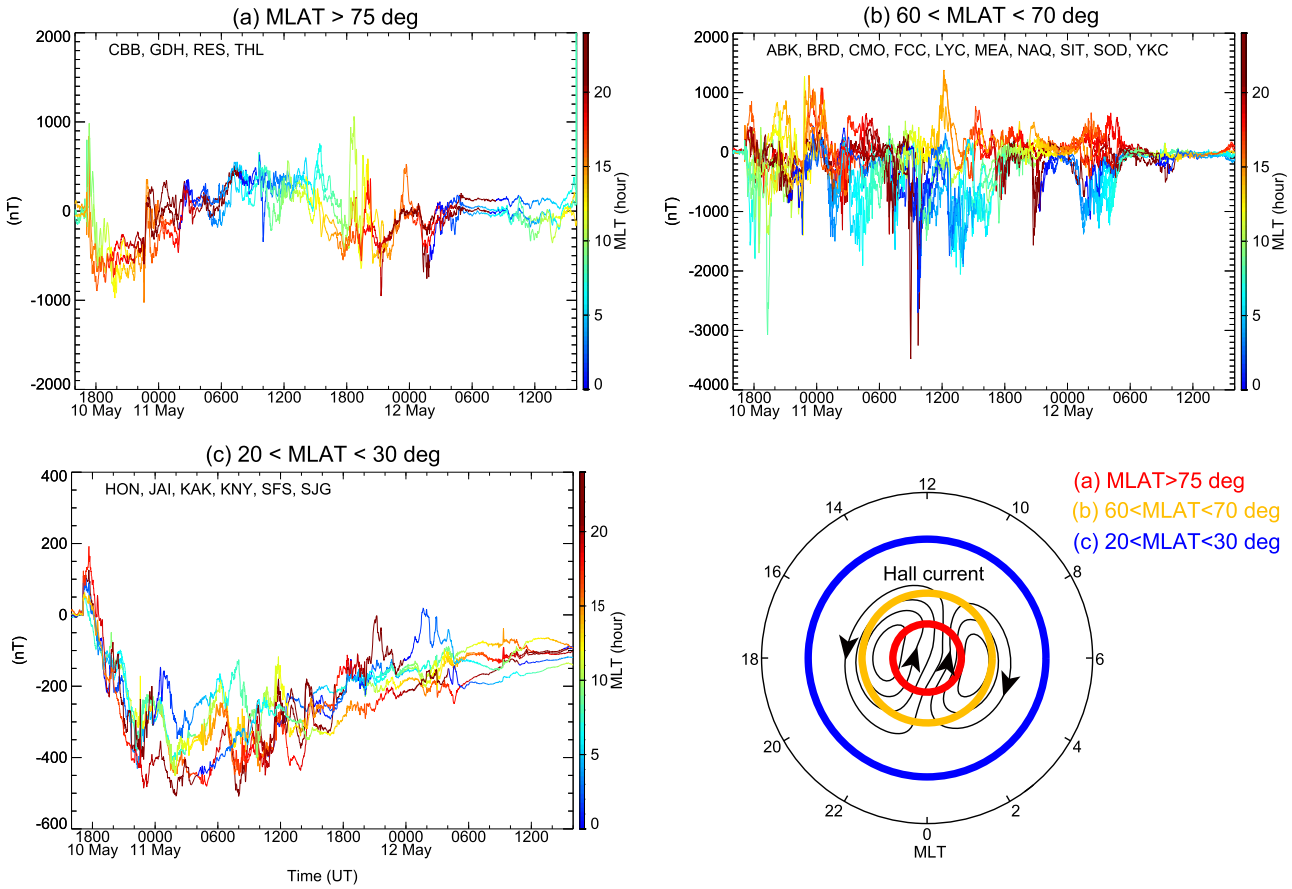


Figure 13. The north component of the geomagnetic disturbances ΔX at (a) MLAT $> 75°$, (b) $60° <$ MLAT $< 70°$ and (c) $20° <$ MLAT $< 30°$, where MLAT stands for the magnetic latitude. The color codes indicate MLT. The abbreviation codes indicate the sites used to plot.

Table 3
Intervals in which IMF B_z Remained Negative for 30 Minutes and More

	Start Time	End Time	Duration (minutes)	Minimum IMF B_z (nT)
1	19:41 UT on May 10	22:23 UT on May 10	162	−40
2	22:41 UT on May 10	01:36 UT on May 11	175	−56
3	01:45 UT on May 11	04:23 UT on May 11	158	−35
4	06:07 UT on May 11	07:37 UT on May 11	90	−40
5	07:57 UT on May 11	11:01 UT on May 11	184	−41
6	11:29 UT on May 11	17:39 UT on May 11	360	−25
7	20:18 UT on May 11	20:48 UT on May 11	30	−9
8	23:06 UT on May 11	23:49 UT on May 11	43	−7
9	00:12 UT on May 12	02:48 UT on May 12	156	−5

Note. IMF B_z is obtained by the 1 minute OMNI data.

data sets for auroral visibility during this storm (e.g., M. Grandin & ARCTICS Collaboration 2024; C. Wicklund 2024) with further possible equatorward extensions of the instrumental detections of the auroral emissions, while they do not separate naked-eye visibility from camera visibility. We located their geographic coordinates and computed their magnetic latitudes using the coordinates of the northern geomagnetic pole (N80.°8 W72.°6) in the IGRF-13 model (P. Alken et al. 2021).

On this basis, we confirmed the equatorward extent of naked-eye auroral visibility in both hemispheres. In the Southern Hemisphere, naked-eye auroral reports have been confirmed down to Boulouparis in New Caledonia (S21°54′, E165°59′, −26.°5 MLAT), Chewko Lookout in Australia (S17°03′, E145° 22′, −24.°2 MLAT), and Tivoli Astrofarm in Namibia (S23°28′, E018°01′, −23.°2 MLAT). In the Northern Hemisphere, naked-eye auroral reports have been confirmed down to El Peyote (N22°41′, W100°36′, 30.°7 MLAT) and Hanle in India (N32° 28′, E078°34′, 24.°3 MLAT).

Auroral visibility at any site does not immediately indicate overhead auroral visibility. Therefore, we must reconstruct the equatorward boundaries of the auroral oval considering the MLAT and the reported altitude at each site. The altitude profiles were used to reconstruct the equatorward boundaries of the auroral oval conservatively. We assumed the altitude of the visual aurorae to be ≈ 400 km, following F. E. Roach et al. (1960) and Y. Ebihara et al. (2017), to keep consistency with

Table 4
Geomagnetic Observatory Sites Used to Plot Figure 13

Abbreviation Code	Geographic Latitude (°)	Geographic Longitude (deg)
(a) MLAT > 75°		
CBB	69.12	254.97
GDH	69.25	306.47
RES	74.69	265.11
THL	77.47	290.77
(b) 60° < MLAT < 70°		
ABK	68.4	18.8
BRD	49.87	260.03
CMO	64.87	212.14
FCC	58.76	265.91
LYC	64.6	18.7
MEA	54.62	246.65
NAQ	61.16	314.56
SIT	57.06	224.68
SOD	67.4	26.6
YKC	62.48	245.52
(c) 20° < MLAT < 30°		
HON	21.32	202.0
JAI	26.92	75.8
KAK	36.23	140.19
KNY	31.42	130.88
SFS	36.7	354.1
SJG	18.11	293.85

the previous estimates for the historical storms (F. Miyake et al. 2019; I. Usoskin et al. 2023). We quantified the equatorward boundaries of the auroral emission regions by measuring their invariant latitude (ILAT) and tracing the footprints of the magnetic field lines where the auroral electrons precipitated, following C. E. McIlwain et al. (1961).

In the Southern Hemisphere, we have reconstructed the equatorward boundaries of the auroral oval down to 29.°8 ILAT based on a Namibian report from Tivoli Astrofarm (S23°28′, E018°01′, −23.°2 MLAT) with a naked-eye auroral visibility up to 45°. This is somewhat followed by the Australian report from Bromfield Swamp (S17°22′, E145°33′, −24.°5 MLAT) with a naked-eye auroral visibility up to 20° and the New Caledonian report from Boulouparis (−26.°5 MLAT) with naked-eye auroral visibility of up to 20° that allow the reconstruction of the equatorward boundaries of the auroral oval down to 35.°0 ILAT and 36.°7 ILAT, respectively. In the Northern Hemisphere, a Mexican report from Chirimoyos (N23°26′, W105°48′, 31.°0 MLAT) allowed us to reconstruct the equatorward boundaries of the auroral oval down to 35.°5 ILAT based on the reported naked-eye auroral altitude of 60°. This is slightly more equatorward than what we can reconstruct from another Mexican report from El Peyote (N22°41′, W100° 36′, 30.°7 ILAT) with a reported naked-eye auroral altitude of 25°. Overall, our reconstructions in both hemispheres were reasonably consistent: 29.°8 ILAT in the Southern Hemisphere and 35.°5 ILAT in the Northern Hemisphere.

Caveats must be noted on these estimates here. First, their naked-eye visibility is considerably different from the camera visibility (Appendix A). Second, SAR arcs often reach farther toward the magnetic equator from the auroral ovals. It is certainly tempting to associate this considerable visibility stretch with the SAR arcs (L. Spogli et al. 2024), whereas this association conflicts with the analysis of the GOLD imagers (D. K. Karan et al. 2024).

The SAR arcs appear monochromatically in reddish coloration, in contrast to the actual reddish auroral emissions (J. U. Kozyra et al. 1997; K. Shiokawa et al. 1997). M. Mendillo et al. (2016) reported that patches of emission at the oxygen green line (557.7 nm) are superimposed to the SAR arcs. After considering the possibility that precipitating ions gave rise to the oxygen red line, M. H. Rees and C. S. Deehr (1961) suggested that precipitation of low energy electrons could result in the emission at the oxygen green line, together with head conduction from the plasmapause–ring current interaction region, which could result in the SAR arcs. For now, we cannot exclude the possibility that the monochromatic reddish aurora witnessed at low latitudes was associated with the SAR arcs, as was also the case with historical auroral reports with monochromatic reddish coloration.

Therefore, we need to choose records with nonreddish colors or explicit structures (Appendix B) to rule out SAR arcs and conservatively reconstruct the equatorward boundaries of the auroral oval to be free from contamination of SAR arcs. The reports that we used to reconstruct the equatorward boundaries of the auroral oval in both hemispheres included nonreddish colorations (pinkish and purplish components in Bromfield in Australia, and pinkish components in Chirimoyos in Mexico; see Appendix B) and/or explicit structures (in Chirimoyos in Mexico and Tivoli Astrofarm in Namibia), as partially seen in Figure 14. Our reconstructions of the equatorward boundary of the auroral oval support analyses of the GOLD imager that suggested auroral stretch down to the southern tip of South Africa and South America (D. K. Karan et al. 2024), rather than the SAR arc hypothesis based on a model calculation (L. Spogli et al. 2024). Therefore, our reconstructions enhance the reliability of an extension of the equatorward boundary of the auroral oval during the 2024 May storm in comparison to the modeling results of L. Spogli et al. (2024).

8. Temporal and Spatial Variation of Electron Density (Total Electron Content) in the Ionosphere

Previously in this paper we elucidated the characteristics of temporal and spatial variations of the geomagnetic field, ionospheric convection, and location of the auroral oval during a geomagnetic storm. Considering the magnitude of geomagnetic field perturbations and ionospheric convection, a substantial electromagnetic energy transfer from the magnetosphere to the ionosphere is expected to produce severe positive and negative storms that cause the electron density in the ionosphere to increase or decrease. To confirm these signatures in the ionosphere during a geomagnetic storm, we analyzed the global total electron content (TEC) data at a high spatial resolution of 0.°5 × 0.°5 (in geographic latitude × longitude).

Figure 16 shows the polar map of the ratio of the TEC difference (rTEC; A. Shinbori et al. 2019) in the Northern Hemisphere in altitude adjustment-corrected geomagnetic (AACGM) coordinates (Section 4.2 in K. M. Laundal & A. D. Richmond 2017) at 17:00 and 22:00 UT on May 10 and 18:00 UT on May 11, corresponding to the prestorm phase, main, and recovery phases of this geomagnetic storm, respectively. During the prestorm phase, the rTEC value was nearly zero across the entire region, except for the west coast of

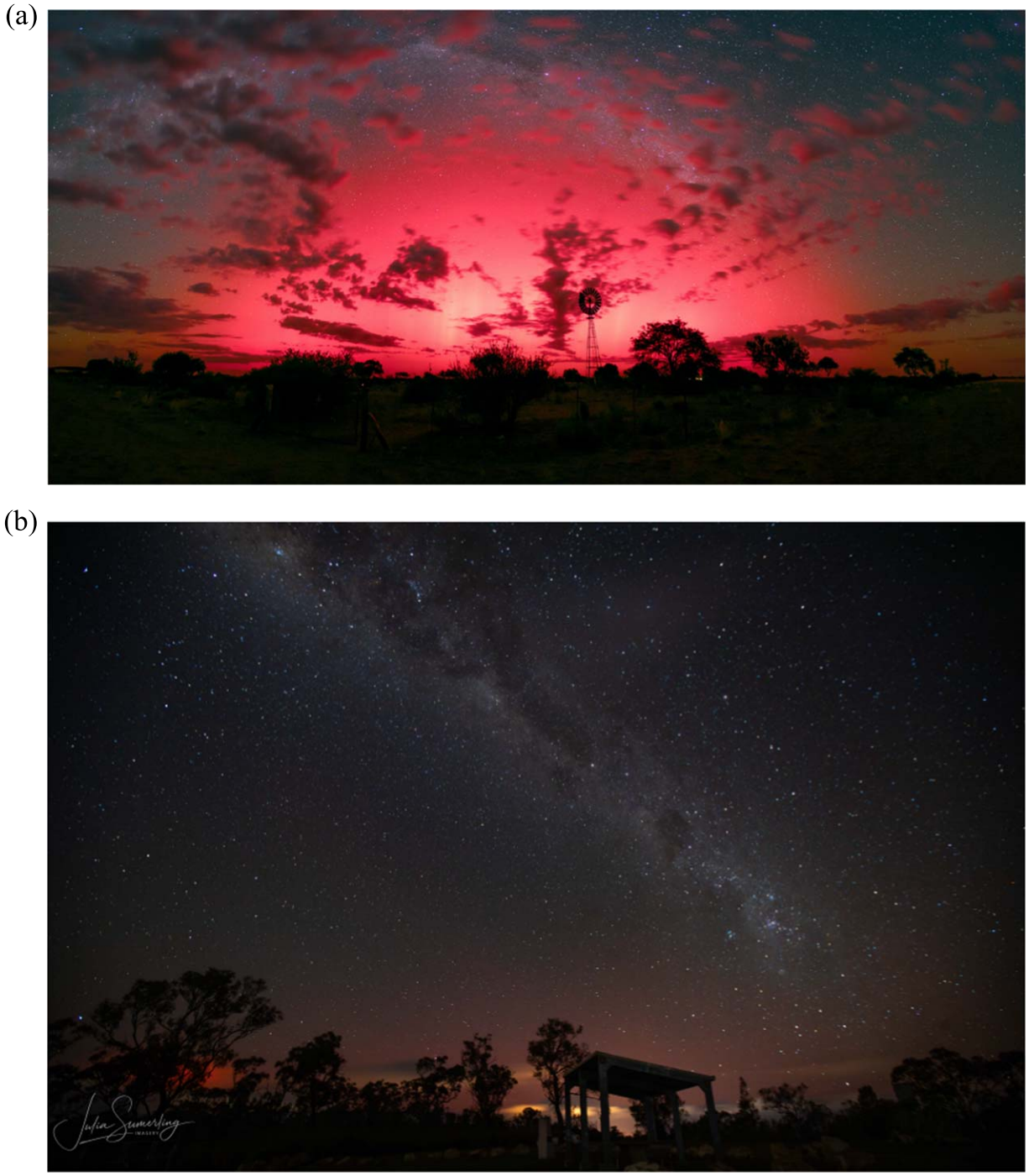


Figure 14. Examples of the reported auroral visibility throughout the world, from top to bottom: (a) Tivoli Astrofarm in Namibia (© Richard Payne), (b) Chewko Lookout near Mareeba in Australia (© Julia Sumerling), (c) Las Grutas in Argentina (© Denis Martínez), and (d) Mogán in Gran Canaria (© Víctor R. Ruiz).

North America, exhibiting an MLAT range of 50°–70°. This means that the spatial distribution of rTEC is geomagnetically quiet. During the main phase, the characteristic structure of rTEC appeared in the high- to low-latitude regions. Furthermore, the rTEC considerably increased to >1.0 at low latitudes (20°–50° MLAT) in the dusk sector (13–20 hr MLT). These characteristics correspond to those of storm-enhanced density (SED; J. C. Foster 1993). Furthermore, the SED is associated with a spatially narrow enhancement of the rTEC, extending from the dayside to the nightside in the polar region. This phenomenon is called the tongue of ionization (TOI) and is caused by plasma transportation from the dayside midlatitude ionosphere into the polar cap owing to the enhanced convection electric fields during the main phase of a geomagnetic storm (M. David et al. 2011; J. C. Foster 1989, 1993). A. Shinbori et al. (2022) demonstrated the seasonal dependence of amplitudes of SED and TOI amplitudes as a function of the geomagnetic storm magnitude, and based on a statistical analysis, they found that the SED and TOI amplitudes were maximal in winter and minimal in summer based on a statistical analysis. However, despite the summer season, when most of the northern polar region is sunlit, a large rTEC enhancement

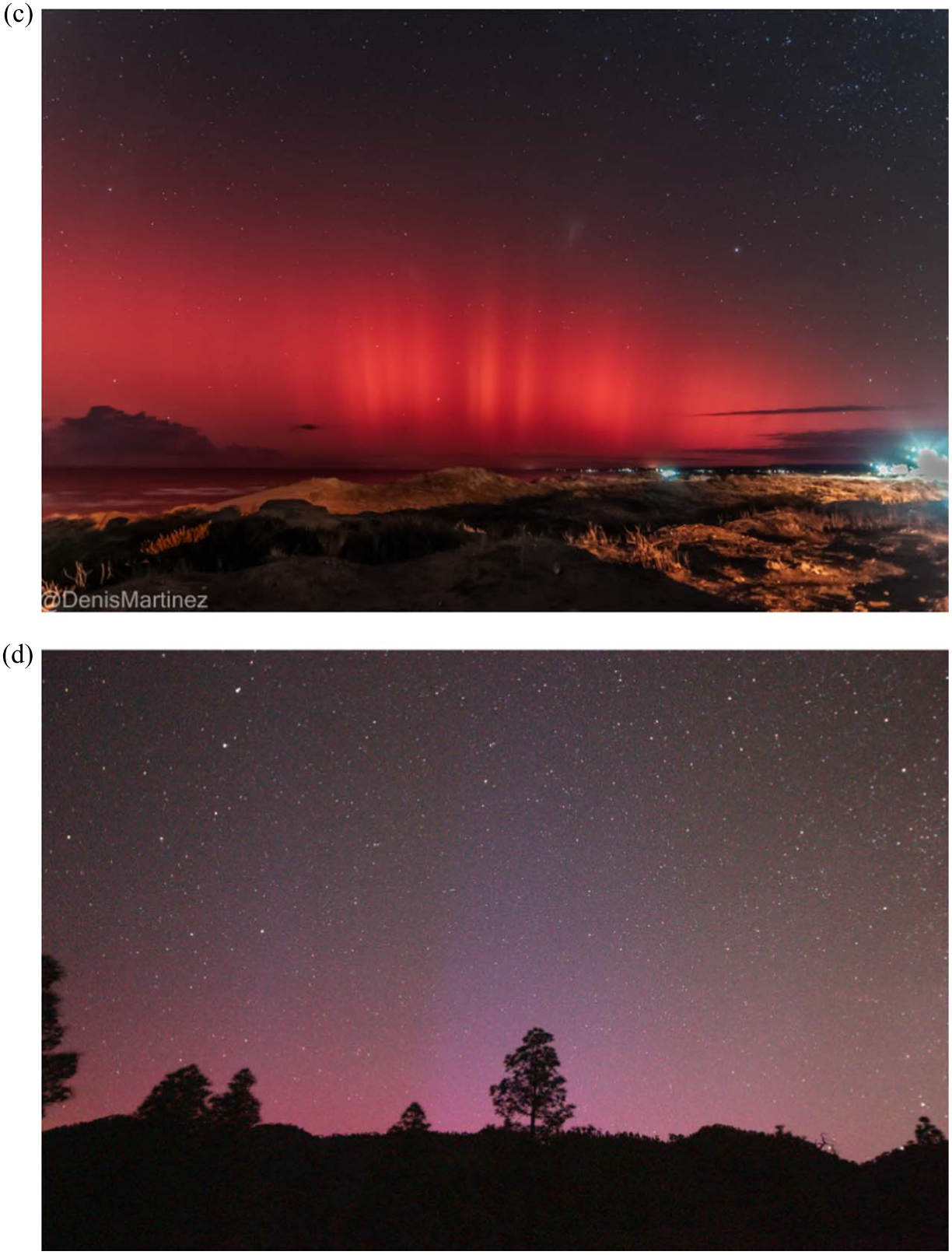


Figure 14. (Continued.)

related to the SED and TOI occurrences during the geomagnetic storms of 2024 May 10–11 was observed, which differed from those reported in previous studies. This implies that an intense convective electric field was imposed from the high-to-low latitudes of the ionosphere during the geomagnetic storm events. Furthermore, a narrowband rTEC enhancement, corresponding to the extension of the auroral oval, was observed in the MLAT range of 45°–55° on the nightside (22 −03 hr MLT), implying that the auroral oval extended to lower latitudes, as described in Section 7. Similar to the rTEC

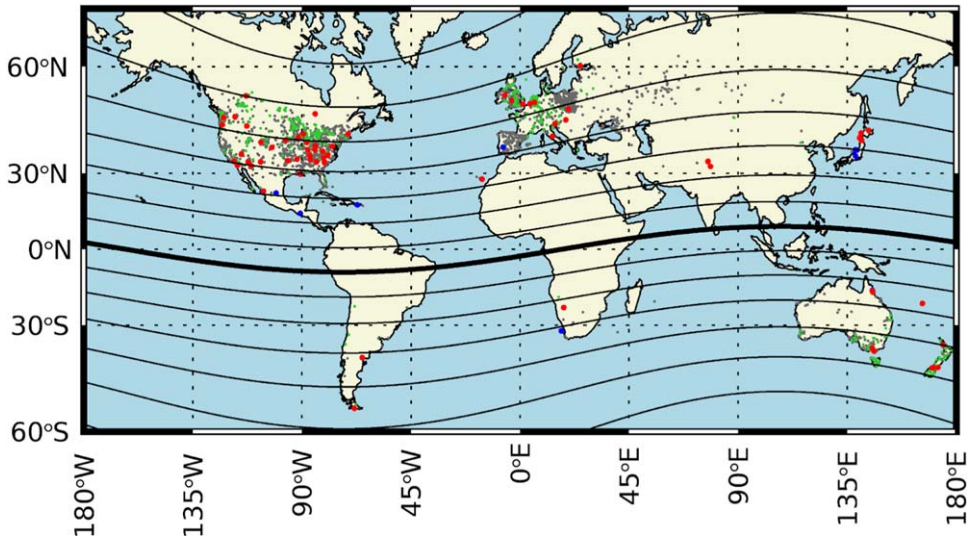


Figure 15. Geographical extents of the reported auroral visibility from 2024 May 10 to 12. The contour lines indicate the magnetic latitude (MLAT) with an interval of each 10°. The solid line indicates the magnetic equator (0° MLAT). The red and blue dots indicated auroral report with naked-eye visibility (this section) and camera visibility only (Appendix A), respectively. For reference, we have plotted visibility reports from M. Grandin & ARCTICS Collaboration (2024) and C. Wicklund (2024) as greenish and gray small dots, respectively.

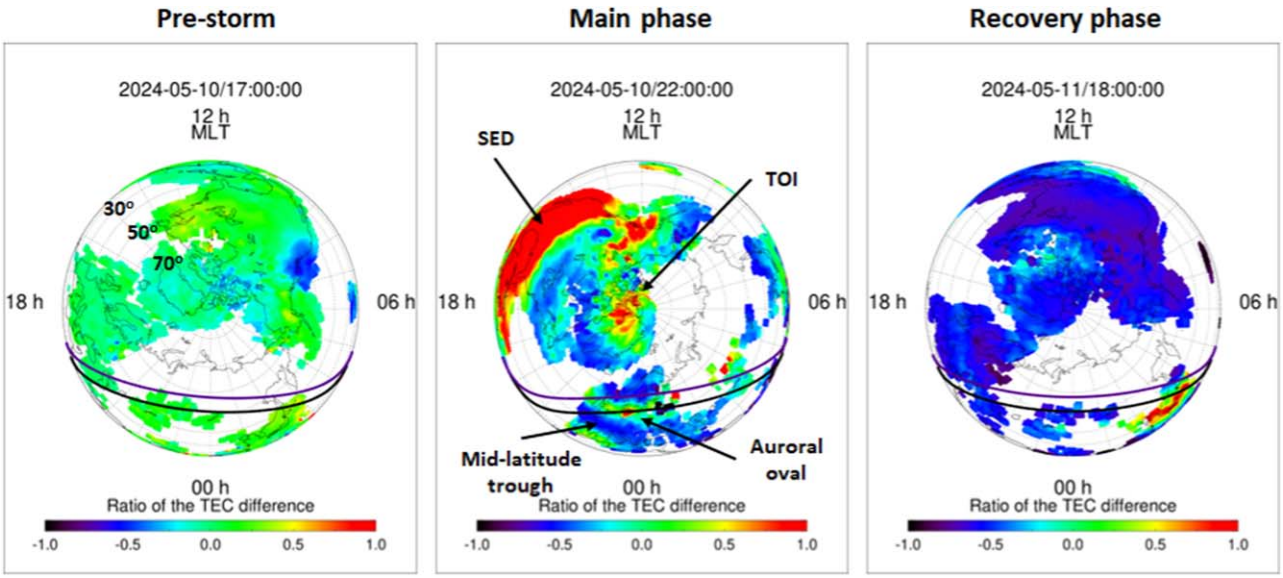


Figure 16. Polar maps of the TEC difference ratio (rTEC) in the Northern Hemisphere represented in altitude adjustment-corrected geomagnetic (AACGM) coordinates during the prestorm, main, and recovery phases. The color bars indicate the rTEC from −1.0 to 1.0. The black and purple curves represent sunset terminators at altitudes of 105 and 300 km, respectively.

enhancement, a narrowband region with decreased rTEC appeared in the MLAT range of 25°–45°. This rTEC depletion represents a midlatitude trough corresponding to the plasmapause location in the inner magnetosphere (A. Shinbori et al. 2019). During the recovery phase, the global spatial distribution of the rTEC considerably decreased in the entire region, from high-to-low latitudes, reaching −0.75 on the dayside. This rTEC depletion corresponded to a negative storm caused by a decreasing electron density in the ionosphere owing to the enhancement of the recombination process associated with an increasing N_2 density. Thus, the extensive depletion of electron density in the ionosphere severely influences the propagation of the electromagnetic waves employed in satellite navigation and positioning.

9. Summary and Discussion

This paper presented a flash report that reviewed the data from the 2024 May solar and geomagnetic storms. The area of the source active region, NOAA AR 13664, rapidly increased from 113 to 2761 MSH from May 1 to 9 (Figures 1 and 2). This rapid growth led to an increase in the total energy of the nonlinear force-free field, potential magnetic field, and free magnetic energy, especially from May 7 onward, producing 12 X-class flares up to X8.7 between May 8 and 15 (Table 1 and Figure 2).

This AR was associated with at least 10 halo CMEs. By evaluating their velocities, we could associate the CME launched at 22:36 UT on 2024 May 8, with the great geomagnetic storm that originated from the SSC at 17:05 UT on May 10 (Figure 4). Thus, the propagation time of the CME was 42 hr 29 minutes.

This CME collided with at least three preceding CMEs, and all of them stacked up before reaching Earth, leading to the detection of large IPS amplitudes. These signals are associated with the high-density region where fast-propagating CMEs accumulate background solar wind and might support CME–CME interactions in interplanetary space.

One of these solar eruptions triggered a significant Forbush Decrease in cosmic ray intensity and a solar proton event. The CMEs caused a relatively large Forbush Decrease with the magnitude of $\geqslant$15%, as seen in neutron monitors' count rate. Subsequently, both the polar neutron monitors and the GOES-16 satellite registered a notable cosmic ray increase from ≈2:00–3:00 UT (Figure 6). This increase was confirmed to be a GLE and included as GLE #74 in the International GLE Database.

These CME arrivals considerably compressed the magnetosphere. THEMIS measurements confirmed their crossings over the boundary of the magnetosphere at ≈8.24 R_E at 17:05 UT (Figure 7), ≈8.51 R_E at 17:05 UT (Figure 8), ≈5.04 R_E at 19:12 UT (Figure 9), and ≈5.37 R_E at 19:19 UT (Figure 10). Particularly, the observations of THEMIS-E and THEMIS-A during their crossings at 19:12 and 19:19 UT, respectively, indicated that the magnetopause underwent a significant inward motion beyond the geosynchronous orbit (6.6 R_E). Additionally, major loss processes for radiation belt particles occur during such great events (D. L. Turner et al. 2012). Moreover, these events have profound space weather implications for geospace satellites because they can be exposed to energetic radiation belt particles that may partially or completely damage their electronic systems (A. D. P. Hands et al. 2018). For example, if a satellite reenters the magnetosphere and radiation belts when the magnetopause moves back out, the likelihood of damage is amplified if energetic particle injections associated with rapid energetic particle injections and wave–particle interactions increase (D. N. Baker et al. 2017; H. C. Koons & J. F. Fennell 2006). Therefore, satellites in geospace may have been exposed to harsh space weather conditions during the geomagnetic storm of 2024 May 10.

The SSC at 17:05 UT upon shock arrival was observed at multiple observatories, such as Kakioka (≈78 nT) and San Fernando (≈130 nT), indicating local enhancements in the duskside, especially in the summer hemisphere. This LT effect probably enhanced SSCs in the European sector.

The resultant geomagnetic storm developed to minimum provisional Dst and Dxt index values of −412 and −415.5 nT, respectively, at 2:00 UT on May 11. In the provisional AE and AL indices, the auroral electrojet was intensified up to 4098 and −3797 nT during the storm main phase, respectively, and 3982 and −3797 nT during the storm recovery phase, respectively. However, the intensity in both indices reduced between the phases, probably because the auroral oval moved equatorward and remained away from the AE/AL observatories. However, their magnitudes should not be exaggerated as statistical studies have indicated that the return period of the Dst index is 5.7 yr (K. Tsubouchi & Y. Omura 2007) and that of the AE index 1.49 yr (M. Nakamura et al. 2015). It is possible that geomagnetic activity during Solar Cycle 24 was considerably lower than that during normal solar cycles.

Furthermore, the DP2 ionospheric current was significantly enhanced (Figure 14). This enhancement manifested as an intensification of the magnetospheric convection, which is highly likely to cause earthward penetration of the hot plasma originating in the plasma sheet, thereby enhancing the ring current.

This storm also induced considerable disturbances in the ionosphere (Figure 16), leading to an SED in the ionosphere, large rTECs in the lower MLATs (20°–50°) on the duskside (13–20 MLT), and tongues of the ionosphere extending from the dayside to the nightside in the northern polar region, despite unfavorable seasonality. These findings indicated the occurrence of an intense convective electric field in the high-to-low MLATs in the ionosphere. These TEC records also confirmed rTEC enhancements in a narrow MLAT band of 45°–55° on the nightside, confirming the equatorward extension of the auroral oval in accordance with the visual records. The rTEC measurements also revealed a plasmapause in the inner magnetosphere, characterized by a depletion in the 25°–45° MLAT region and a considerably negative storm globally during the recovery phase.

Among the records collected since the beginning of the space age (1957 onward), the min Dst of −412 nT is comparable to those of four of the five greatest geomagnetic storms, specifically those that occurred in 1957 September (min Dst = −427 nT), 1958 February (min Dst = −426 nT), 1959 July (min Dst = −429 nT), and 2003 November (min Dst = −422 nT), and significantly more moderate only than that of the extreme storm recorded in 1989 March (WDC for Geomagnetism at Kyoto et al. 2015a; P. Riley et al. 2018). This value should be revised in the future because this provisional Dst index was developed using a method different from that for the final Dst index that extended only up to 2016 at the time of writing this manuscript. Additionally, this storm recorded the highest value on the *Kp* index (*Kp* = 9) (J. Matzka et al. 2021) since 2003 October (Y. Yamazaki et al. 2024). Moreover, it is the second to the fifth greatest on the Hp30 index since 1985 (Y. Yamazaki et al. 2024). In this regard, this geomagnetic storm was certainly large but not unique.

During this storm, the auroral display was visible to the naked eye down to |23.2|° MLAT (Tivoli Astrofarm in Namibia; Figure 14). This allowed us to reconstruct the equatorward boundaries of the auroral oval at an ILAT of 29°.8. In Figure 17, this storm is contextualized with the auroral visibilities of historical storms, as modified from the previous presentations in D. J. Knipp et al. (2021) and H. Hayakawa et al. (2024). Based on naked-eye observations, this storm was comparable to the great storms of 1958 February and 1959 July, for which the equatorward boundary of the auroral oval was reconstructed down to |23.3|° and |27.4|° MLAT, respectively (H. Hayakawa et al. 2023c, 2024).

Historical records indicate that the equatorward boundaries of auroral ovals have extended down to 33°–40° ILAT during the greatest geomagnetic storms of the space age (D. H. Boteler 2019, H. Hayakawa et al. 2023c, 2024) and 24°–25° during extreme storms in history (H. Hayakawa et al. 2020, 2023b). Analyzing modern naked-eye observations of auroral ovals during geomagnetic storms and comparing them with those of historical ones is crucial for better contextualizing great storms.

However, our reconstruction was most likely an ultraconservative estimate. The scientific community hosts large-scale citizen science portals (E. A. Macdonald et al. 2015, B. C. Kosar et al. 2018a, C. Wicklund 2024), which can significantly increase the number of auroral records. The utility of these citizen science aurora data sets such as Aurorasaurus has already been confirmed through ground-truth validation of satellite observations and boundary estimations in a case study of the intense geomagnetic storm of 2015 March (B. C. Kosar et al. 2018b). In the near future, we will request detailed reports from individual contributors to document the temporal evolution (Figure 12), spatial evolution

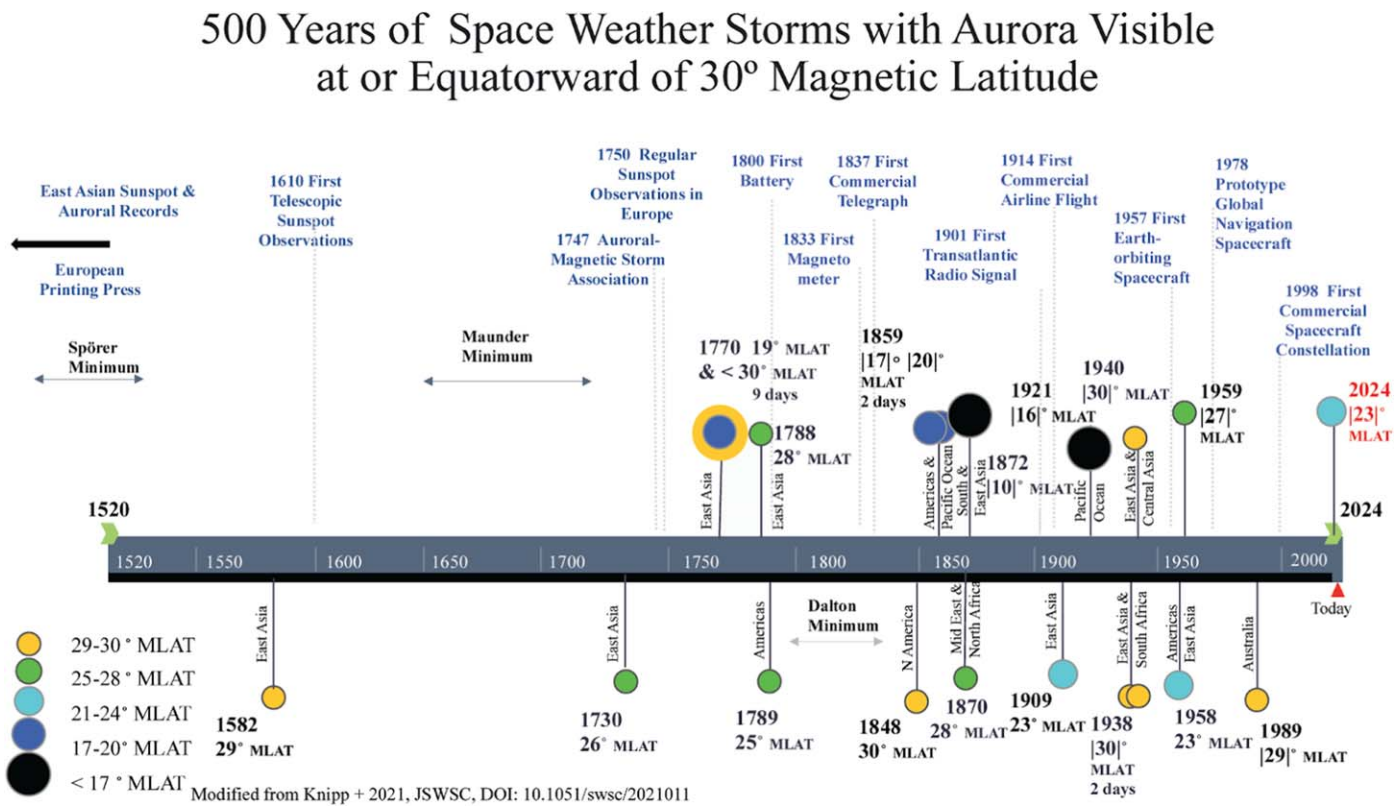


Figure 17. Spatial extents of naked-eye auroral visibility of great historical geomagnetic storms whose auroral visibilities have been reported as <|30|° MLAT. This figure is a modified version of those presented in D. J. Knipp et al. (2021) and H. Hayakawa et al. (2024).

(Figure 15), and coloration of the reported auroral displays (Figure B1). These details will allow us to enhance the reconstruction of auroral emissions, their temporal/spatial evolution, and their equatorward auroral oval boundaries. Moreover, as this storm occurred in mid-May, the European and North American sectors had the disadvantages of longer daytime durations and the advantages of better weather conditions. The timing of the storm peak suggests that the auroral visibility conditions were favorable in the Southern Hemisphere, particularly in South America. Therefore, we need to request the recollection of this visual aurora by citizens, especially individual astronomers, in multiple languages. This requirement is confirmed with multiple critical reports in Latin America and Middle East (Figure 15), where non-English speakers are the majority of the local populations. Additionally, outreach efforts to explain scientific phenomena in plain language are imperative for including and empowering the public, as exemplified by the Aurorasaurus blog.

Immediately after this storm, auroral photographs were uploaded from various regions worldwide. This significant geographical extent can be partially explained by the differences in the sensitivities of naked-eye observers and cameras. Today, we have access to significantly better optical instruments, such as cameras, than during past great space weather events. Moreover, space weather forecasts have allowed and motivated people to observe aurorae at the right time and direct cameras to the sky to figure out whether they can capture aurorae, as confirmed by several case reports. Furthermore, Internet/SNS accessibility allows observers to swiftly exchange reports/photos, which motivates more people to search for visible aurorae. However, light pollution in urban areas hinders aurora visibility in the night sky (M. Smith 2009; F. Falchi et al. 2016). Therefore, witnesses of this auroral storm have three advantages and one disadvantage when documenting auroral extensions during great geomagnetic storms compared to those witnessed in historical auroral storms.

This study is no more than a flash report conducted using provisional data sets, and more precise data sets must be accumulated and compared with the modeling results. Through further analyses, these solar and geomagnetic storm features may allow us to bridge our modern scientific knowledge with that of historical great solar and geomagnetic storms and better assess space weather extremes.

Acknowledgments

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Data Availability

We have acquired sunspot numbers and sunspot drawings from the SILSO and USET of the Royal Observatory of Belgium, the solar wind parameters and proton and X-ray fluence from the NOAA, the IPS data used from the ISEE, Nagoya University website https://stsw1.isee.nagoya-u.ac.jp/vlist/, neutron monitor data from the International GLE database, and geomagnetic Dst, AE, and AL indices from WDC and the location of north geomagnetic pole for Geomagnetism at Kyoto (https://wdc.kugi.kyoto-u.ac.jp/). We also acknowledge the Oulu Cosmic Ray Station, the Neutron Monitor Database, and individual neutron monitor station managers for maintaining the cosmic ray measurement data sets over a long time period. The TEC data are available on the ISEE, Nagoya University website https://stdb2.isee.nagoya-u.ac.jp/GPS/GPS-TEC/. We thank Samuel Freeland for offering Event Archives of Solar Soft in the Lockheed Martin Solar & Astrophysics Laboratory.

Appendix A
Spatial Extents of the Auroral Visibility and the Auroral Oval: Instrumental Observations

Upon this storm, considerable observers managed to photograph the auroral displays with their instruments, such as digital cameras and smartphones. Such camera observations managed to capture the auroral visibility and auroral oval further equatorward. Good examples can be easily obtained from Japanese cases. As shown in Figure A1, the aurora was photographed at least down to Koumi (N36°04, E138°24, 28°.1 MLAT). More photographs on social media show auroral visibility at lower latitudes, but they require further careful validation owing to fakes, contaminations, and confusions (e.g., P. Meel & D. K. Vishwakarma 2020). In contrast, the naked-eye visibility is confirmed only down to Hiranai (N41°01′, E140°53′, 33°.2 MLAT). The Hiranai photographs captured aurora on the basis of their evident ray structure, while the Koumi photograph might have captured SAR arcs, based on the appearance of monochromatic reddish color (Figure A1).

The Hiranai report showed striking differences in the spatial auroral extents in the naked eye visibility and camera visibility: by the naked eye, up to 20° in elevation, and by optical instruments, up to 70° in elevation. Following the procedure described in Section 7, this study allowed us to reconstruct the equatorward boundary of the auroral oval at Hiranai, down to 43°.0 ILAT and 36°.8 ILAT, for the naked-eye-based and camera-based estimates, respectively. The equatorward boundary looks significantly decent in contrast to our reconstructions in Section 6, as Japanese observers were under broad daylight around the storm peak (≈2 UT), only managed to see the storm recovery phase, and probably missed the most dramatic part of the auroral display.

Similar cases have been reported throughout the world. An Indian auroral report from Hanle in India (N32°28′, E078°34′, 24°.3 MLAT) described the auroral altitude therein as 5° by the naked eye and 20° by camera. Similarly, the New Caledonian report from Boulouparis described auroral altitude as 20° by the naked eye and 30° by a camera. At Tivoli Astrofarm in Namibia (S23°28′, E018°01′, −23°.2 MLAT), aurora was confirmed up to 45° by the naked eye and up to 55° by the camera. Subsequently, its internal structure was confirmed to be up to 20° by photographic analysis (Figure 14). Notably, these examples occurred at different times during the storm, as shown in Figure 12.

Overall, in the Northern Hemisphere, the aurora was photographed as far equatorward as San Juan Sacatepéquez in Guatemala (N14°43′, W090°39′, 23°.4 MLAT) subvisually without naked-eye visibility. In the Southern Hemisphere, the aurora was photographed as far equatorward as Big Mitchell Reserve in Australia (S16°48′, E145°21′, −23°.0 MLAT). This estimate is probably ultraconservative. Independent citizen science data sets accommodate potential auroral photographs from further equatorward, down to São Paulo in Brazil (S23°48′, W046°38′, −15°.2 MLAT) and Lima (S12°05′, W076°52′, −2°.9 MLAT) in the Southern Hemisphere and Ad-Dāḫilīyah in Oman (N22°52′, E057°32′, 16°.8 MLAT) and Calabarzon in the Philippines (N14°06′, E121°05′, 5°.2 MLAT) in the Northern Hemisphere, according to C. Wicklund (2024). However, as C. Wicklund (2024) does not provide further details in his data file, we need to carefully assess their reliability on the basis of their source reports.

Their geographical extensions are significantly different from those in the naked-eye visibility. The instrumental detectability should be of course somewhat variable up to the instrumental setups, such as exposure time. This variation is particularly tricky, as each observer had their own instruments and settings, such as digital cameras and smartphones, without standardizations, unlike well-prepared campaign observations. Such record characteristics have made calibration and homogenization of their photographs extremely difficult.

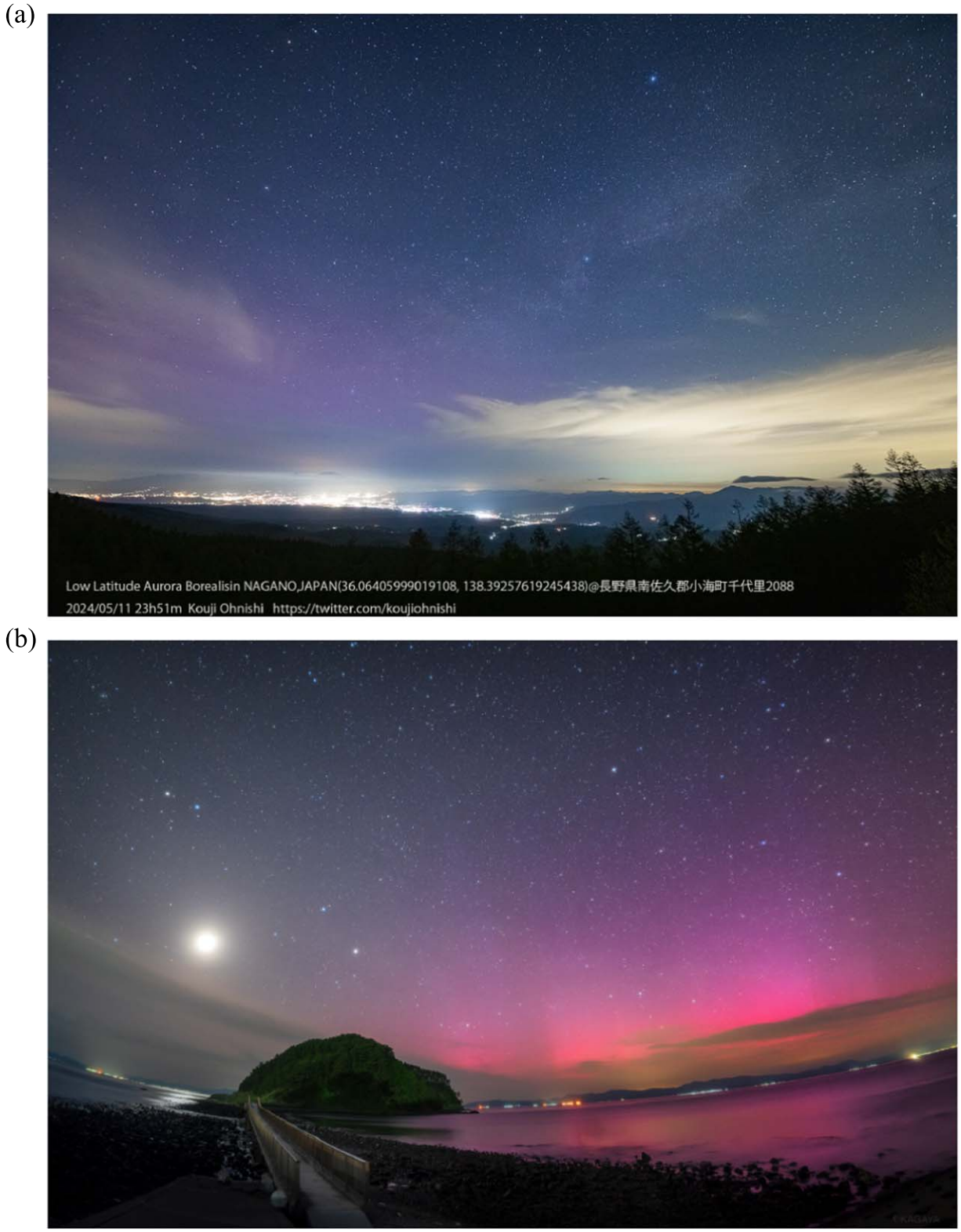


Figure A1. Examples of auroral photographs from Japan, from the top to the bottom: (a) Koumi in Japan with Canon EOS R5 (© Kouji Ohnishi) and (b) Hiranai in Japan with SONY α7R IV (© Yutaka Kagaya). Kouji Ohnishi confirmed that the aurora was not visible to the naked eye. Yutaka Kagaya confirmed the spatial extent of the auroral display up to 20° in elevation by the naked eye and up to 70° in elevation by optical instruments.

The equatorward extent of the visibility with a camera is different from that with the naked eye. This discrepancy is primarily attributed to the different sensitivity. The sensitivity of the camera is much higher than that of the naked eye. We could suggest three possible scenarios for its provenance as detectability differences in (1) vertical extents, (2) latitudinal extents, or (3) their combinations. For Scenario (1), the aurora at high altitudes could have been detected only by the cameras so that the aurora was visible at lower latitudes. For this scenario, the equatorward boundary is assumed to be the same. For Scenario (2), the

equatorward boundary of the auroral oval was fairly broad in latitude, and the aurora extended further equatorward diffusively. According to the DMSP satellite observations, the equatorward boundary of the electron precipitation is broad in latitude during large magnetic storms (K. Shiokawa et al. 1997; Y. Ebihara et al. 2017). Precipitations of electrons with low energy and low flux could cause auroral emission further equatorward, but the auroral emission could be too dim to be visible to the naked-eye observations. For this scenario, the altitude distribution of the aurora is assumed to be the same. It is of course possible that both were the case, as expected in Scenario (3). Further quantitative assessments are needed here.

For the moment, we accommodate as wide error margins as possible. It is empirically known that the maximum brightness of the reddish component of the aurora (at 630.0 nm) is located around 220 km in altitude, and it can range from 200 to 300 km (S. C. Solomon et al. 1988). On the basis of spectra of precipitating electrons observed at midlatitudes near the storm maximum of the large magnetic storm of 1989 March 13–14, the volume emission rate of 630.0 nm peaks at about 270 km altitudes (Y. Ebihara et al. 2017). At 400 km altitude, the volume emission rate of 630.0 nm is an order of magnitude lower than that at the peak altitude. At 500 km altitude, the volume emission rate is much lower than that at the peak altitude. From this consideration, we set the altitudes of the upper limit of the aurora identified by a camera to be 300, 400, 500, and 800 km. In this case, the Namibian report from Tivoli Astrofarm (−23°.2 MLAT), which has a local altitude with instrumental detectability of 55° above the local horizon, allows us to derive the equatorward boundaries of the auroral oval as, at least, 27°.6, 28°.9, 30°.1, and 33°.3 ILAT, respectively. Alternatively, the Australian report from Big Mitchell Reserve (−16°.8 MLAT), which has a local altitude with instrumental detectability of 50° above the local horizon, allows us to derive the equatorward boundaries of the auroral oval as 22°.4, 24°.0, 25°.4, and 29°.1 ILAT, respectively. Further detailed assessments are obviously needed with parallel case studies.

Appendix B
Spatial Distribution of the Auroral Activity

Figure 15 shows a graphical summary of the temporal and spatial evolution of auroral visibility during the 2024 May storm, showing variations in the maximum auroral brightness. Based on the provisional Dst index, the auroral visibility below |40|° MLAT was concentrated in the late main phase and early recovery phase of this storm. One interesting account is a subvisual auroral observation from a ship in the South Atlantic Ocean (S32°00′, E17°00′, 31°.5 MLAT) that lasted from 18:15 to 18:45 UT on May 10. This aurora was photographed shortly after an SSC, despite its occurrence at lower latitudes. A similar case was observed during the 1989 March storm (D. H. Boteler, 2019).

This figure also shows the temporal and spatial evolution of the maximum auroral brightness at each site. We gathered and compared citizen contributors' self-classification and descriptions of the maximum brightness in total illumination with an IBC reference (J. W. Chamberlain, 1961, p. 124) with an extension: (0) not visible by the naked eye, (1) Comparable to Milky Way, (2) comparable to thin moonlit cirrus clouds, (3) comparable to moonlit cumulus clouds, (4) comparable to the full moon, and (5) brighter than any such references. This figure shows that the maximum brightness was higher in higher latitudes and lower at lower latitudes. This, in turn, shows gradual decreases in auroral brightness in lower MLATs and partially explains the smaller geographical extent of the naked-eye auroral visibility than that by instrumental observations.

Figure B1 shows the number of auroral colorations reported in each MLAT range by the naked eye and the camera. A considerable number of reports show reddish and greenish coloration. These colors are typically associated with oxygen emissions at 630.0 nm (reddish) and 557.7 nm (greenish), owing to the precipitation of electrons. Reddish aurorae can more easily be seen than green aurorae by camera. Greenish aurorae cannot be spotted by the eye as far south, generally because they occur at lower altitudes. The yellowish color is most likely caused by a mixture of reddish and greenish colors. The number of reports with greenish coloration was

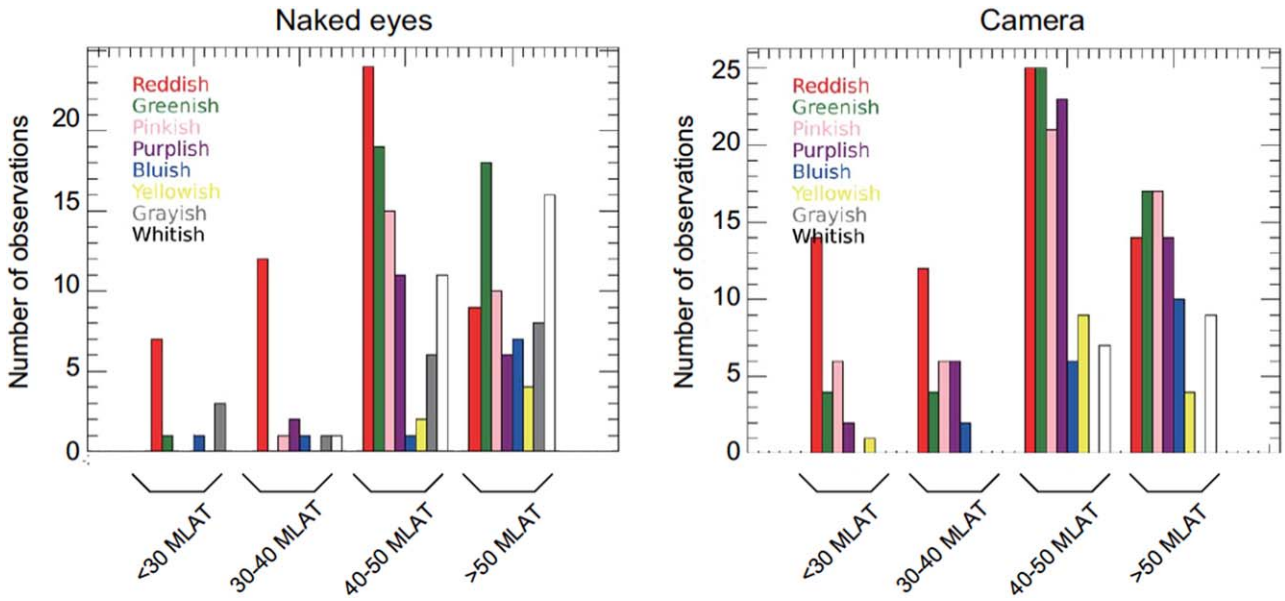


Figure B1. Amounts of reported auroral colorations in each MLAT ranges by the naked eye (left) and by the camera (right), showing (1) <30° MLAT, (2) 30°–40° MLAT, (3) 40°–50° MLAT, and (4) >50° MLAT, from left to right.

comparable to that with reddish coloration at >40° MLATs, whereas it significantly decreased below 40° MLAT by naked-eye observation. For camera observations, a greenish coloration was reported below 40° MLAT. Purplish and pinkish colors have also been reported below 40° MLAT. They are typically associated with nitrogen emissions, including 427.8 nm for N_2^+. Although bands of N_2^+ are known to coexist in the reddish auroral arc (M. H. Rees & C. S. Deehr 1961), this may challenge the conventional wisdom that a low-latitude aurora is dominated by a reddish color. For the purplish auroras that appear at high altitudes, the energy of the precipitating electrons is not necessarily high. The coloration tempts us to associate them with sunlit aurorae (D. M. Hunten 2003), based on the similarity with the past auroral displays with bluish/purplish coloration during the great geomagnetic storms in 1909 and 1958 (H. Hayakawa et al. 2019, 2022). The purplish aurora appeared to be dominant over the reddish one in some of the pictures, as exemplified by Víctor R. Ruiz's photograph in Mogán of Gran Canaria (Figure 14). The dominance of the purplish aurorae should be investigated in future studies, including different response bands of cameras and cell phones. It should be noted that the greenish coloration should have been seen in different colors, such as greyish, by naked-eye observation.

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